# Reference Textbook



# UV-Visible Spectrophotometry

Ultraviolet-Visible Spectrophotometry

Chapter 8 · Jiang Ming

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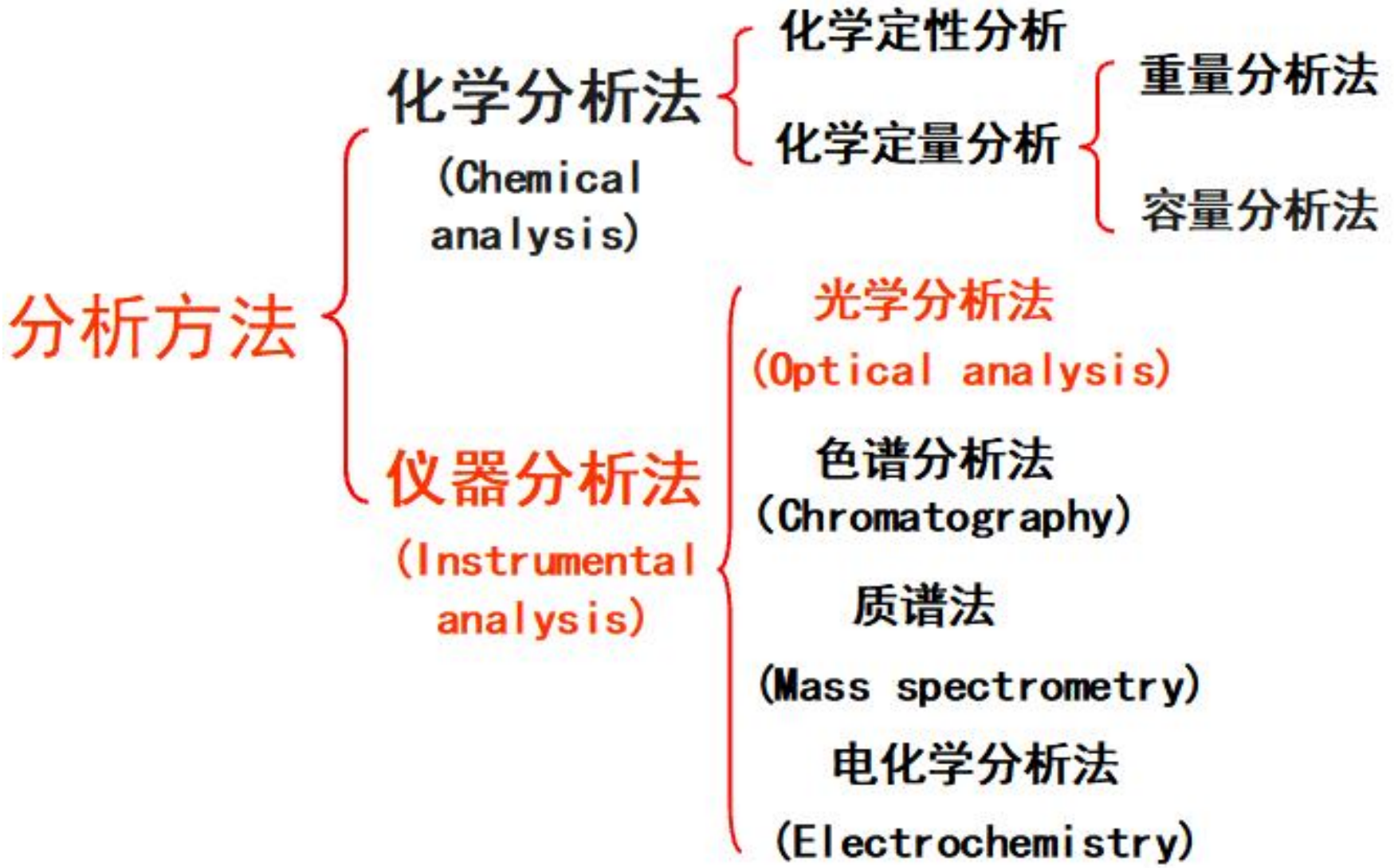
§5 Qualitative Analysis Methods

§6 Quantitative Analysis Methods

# 01

# Overview

# Classification of Analytical Methods Instrumental

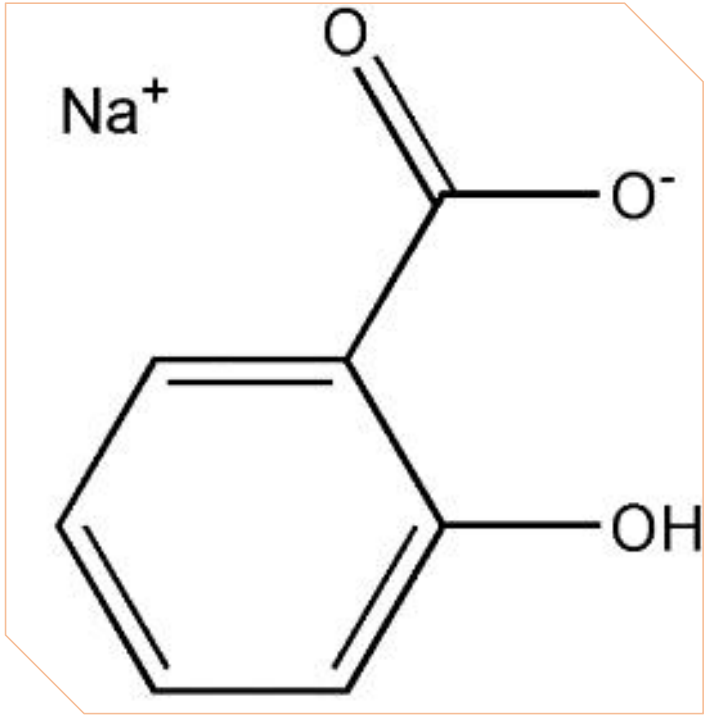


# In-class Question (1/3): Determination of Sodium Salicylate

- The structure of sodium salicylate contains a benzene ring and a carboxylate group

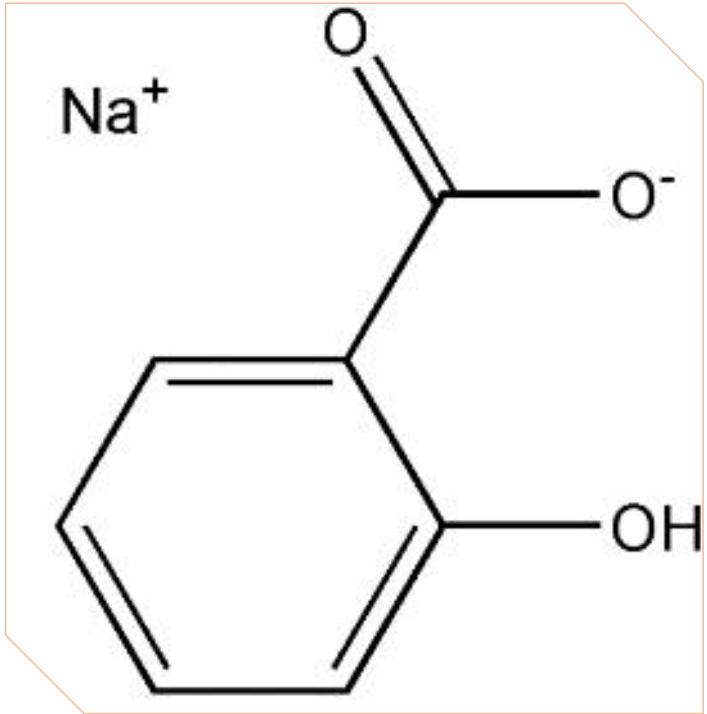
- How can chemical analysis determine its content?

- Hint: sodium salicylate is a very weak base in water ($K_b \approx 5.6\times10^{-10}$)



# In-class Question (2/3): Limits of Non-aqueous Titration

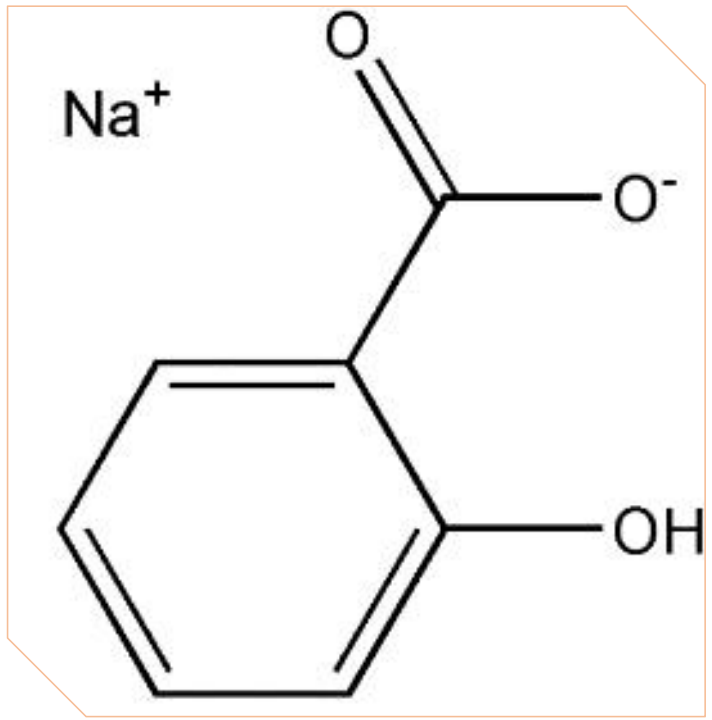
- Non-aqueous titration is still a chemical method
- Sample amount, sensitivity, analysis speed?
- Can complex mixtures be separated, identified or structurally determined?
- Equipment requirements and automation?



# In-class Question (3/3): Choice of Method

- ~100 µg/L in pure water →UV-Vis spectrophotometry

~100 ng/L in blood → high-sensitivity method plus separation technology



# Features of Instrumental Analysis

- Advantages:
- 1. Small sample amount;
- 2. High sensitivity; much lower detection limit;
- 3. Fast; easy to automate and monitor online;
- 4. Can separate, identify or determine structures in mixtures;
- Disadvantages:
- Expensive equipment

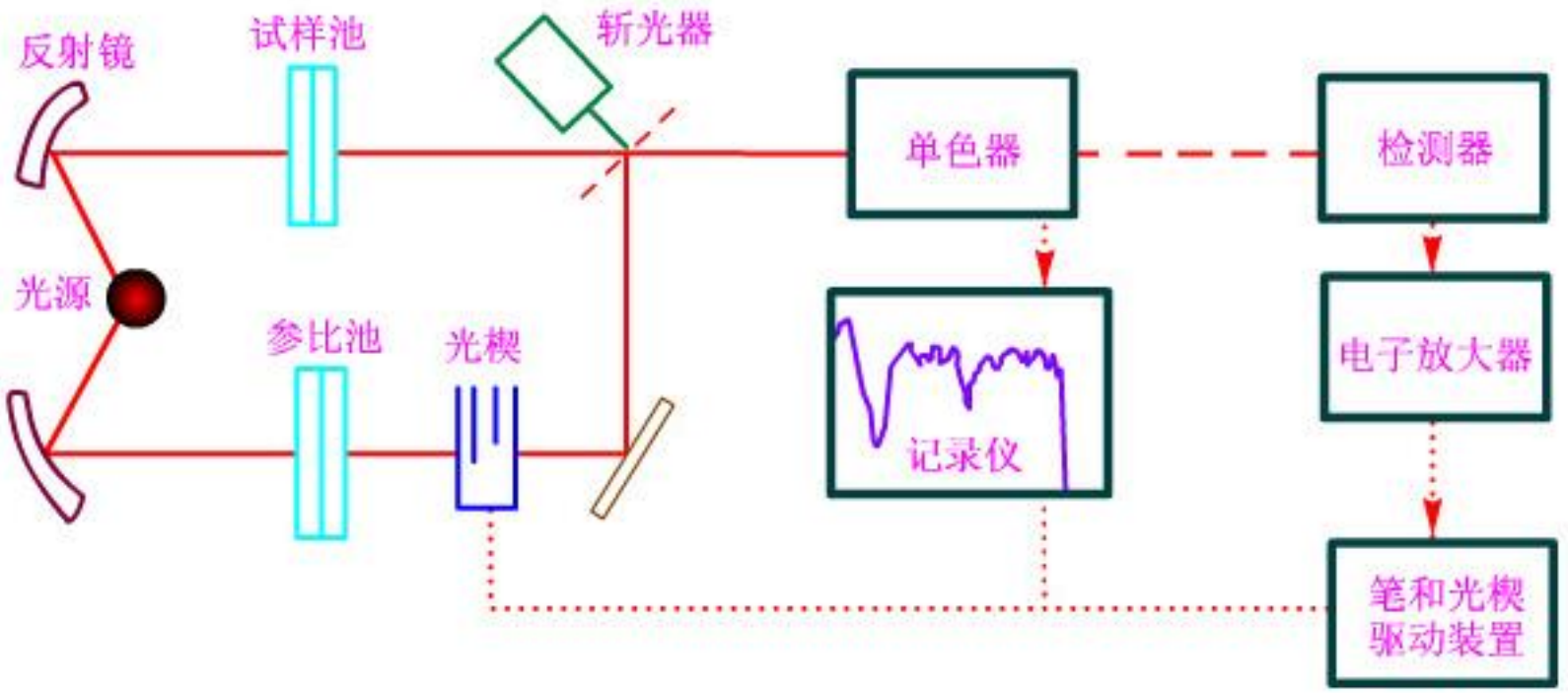
# Reference Prices of Some Instruments

- UV: ~200k CNY
- IR: ~200–400k CNY
- Fluorescence: ~200k CNY
- GC: ~300k CNY
- HPLC: 400k CNY
- MS: 600k–4M CNY
- NMR: 2M–12M CNY

# Optical Analysis Methods:

- Instrumental methods that determine the composition, content and structure of substances based on radiation emitted by the substance, or radiation/signals produced or changed by interaction between the substance and radiation, are collectively called optical analysis methods.

# Basic Features:




- 1. Selective measurement, no mixture separation (unlike chromatography);
- 2. Involves many optical components.

02

# Basic Principles of UV-Vis Spectrophotometry

Fundamental Principles

# UV-Visible Spectrophotometry

ultraviolet-visible spectrophotometry;(UV-Vis)

- A method in which molecules of certain substances absorb radiation in the 200–760 nm region for analysis.

# Features:

① High sensitivity, generally $10^{-4}$~$10^{-7}$ g/mL; suitable for micro/trace analysis.

② High accuracv: relative error 1%~3%.

| | 滴定分析法 | 紫外分光光度法 |
|---|---|---|
| 典型浓度水平 | 常量（如 0.1 mol/L） | 微量（如 $10^{-5}$ mol/L） |
| 相对误差 | 约 ±0.2% | 约 1%~3% |
| **绝对误差** | 相对较大 | **非常小** |

③ Simple and fast operation.

④ Wide range of applications

# Basic Principles of UV-Vis Spectrophotometry

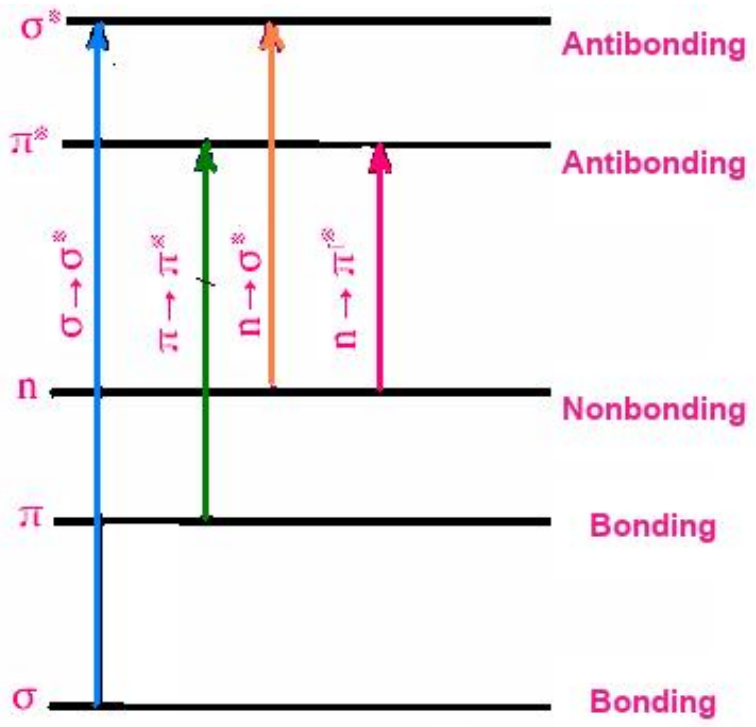
- I. Types of Electronic Transitions
- II. Common Terms
- III. Absorption Bands and Molecular Structure
- IV. Factors Affecting Absorption Bands
- V. Lambert-Beer Law
- VI. Deviation from the Beer Law

# I. Types of Electronic Transitions

- UV absorption spectra arise from transitions of valence electrons in molecules.There are several different types of valence electrons in organic molecules:

- Electrons forming single bonds are called σ electrons;
- Electrons forming double bonds are called π electrons;
- Lone-pair electrons on heteroatoms (O, N, S, X) not involved in bonding are called n electrons.

# Relative Energy Levels of Valence Electrons and Transitions




**σ→σ* > n→σ* ≥π→π*> n→π***

**π→σ*  σ→π* transitions have lower probability**

# 1. σ→σ* Transition

| Compound | λ(nm) | Intensity/ε | transition |
|---|---|---|---|
| $CH_4$ | 122 | intense | σ-σ*(C-H) |
| $CH_3CH_3$ | 130 | intense | σ–σ* (C-C) |

• Saturated hydrocarbons have only low-energy σ bonds; the transition needs large energy and gives strong absorption. Their peaks are generally below 150 nm, beyond ordinary UV range.

# 2. π→π* Transition

• For π electrons of isolated unsaturated double bonds, π→π* transitions occur at ~160–200 nm with very large ε (generally $\varepsilon > 10^4$): strong absorption.

Examples:

• Carbonyl C=O: λ≈188 nm in n-hexane, ε≈900

• Alkene C=C: λ≈177 nm in n-heptane, ε≈13000

• Alkyne C≡C: λ≈178 nm in n-heptane, ε≈10000

# 3. n→π* Transition

• Compounds with heteroatom unsaturated groups (C=O, C=S, -N=N-, etc.) excite n electrons to π orbitals: near-UV (200–400 nm), small ε (10–30, up to 300), weak absorption.

Examples:

- Carbonyl C=O: λ≈280 nm, ε≈16
- Carboxyl –COOH: λ≈204 nm in ethanol, ε≈41
- Nitro $–NO_2$ : λ≈280 nm, ε≈22

# 4. n→σ* Transition

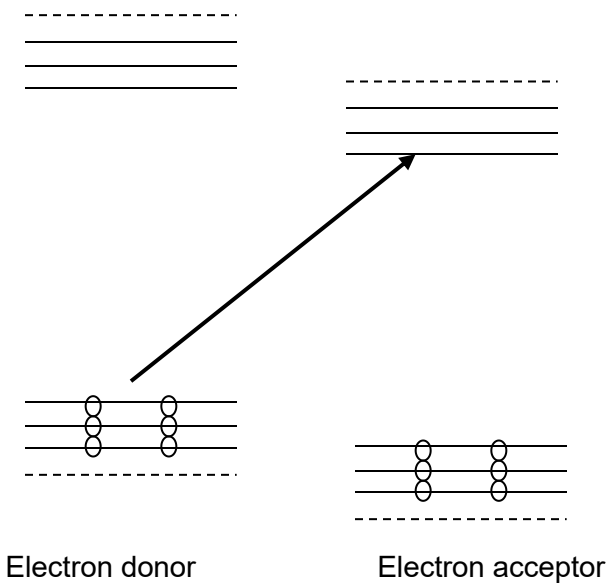
- Saturated compounds with –OH, –$NH_2$ , –X, –S etc. give n→σ* transitions at ~200 nm with weak absorption.

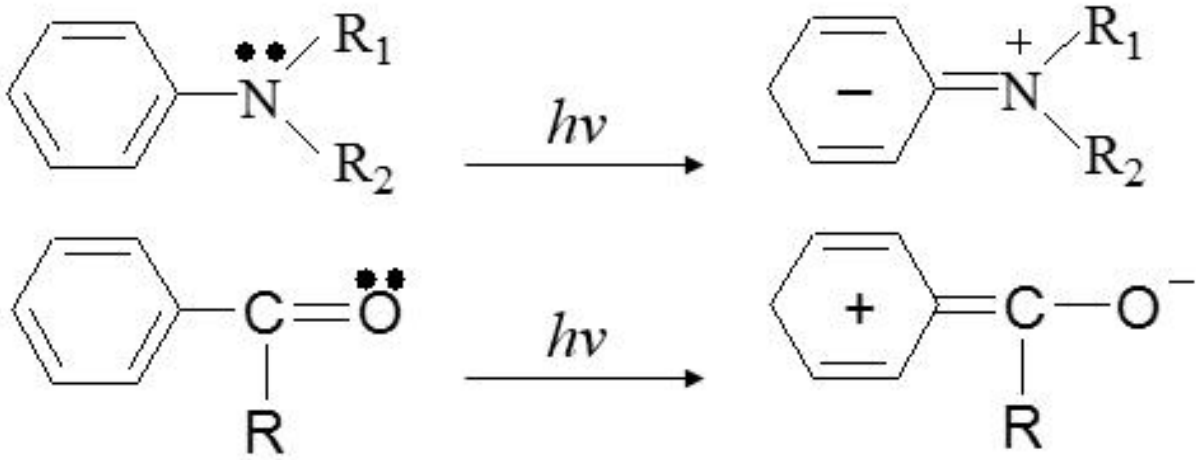
Examples:

- C–O:$CH_3OH$ λ≈183 nm, ε≈200
- C–S:$CH_3SH$ λ≈235 nm, ε≈180
- C–N:$CH_3NH_2$ λ≈210 nm, ε≈300
- C–I:$CH_3I$ λ≈258 nm, ε≈380

# 5. Charge Transfer Transition

• In molecules that are both electron donors and acceptors, radiation excites electrons from donor to acceptor (intramolecular redox), giving strong absorption: a charge transfer spectrum.



# 5. Charge Transfer Transition

## Inorganic compounds

Upon absorbing UV-Vis radiation, charge localized on metal M orbitals transfers to ligand L orbitals (or in reverse). This is a charge transfer transition and the resulting spectrum a charge transfer spectrum.

- **Example: $Fe^{3+}$ and $SCN^-$ form a blood-red complex with a strong peak at 490 nm, essentially via the following reaction:**

**$[Fe^{3+}\ SCN^-]^{2+} + h\nu = [Fe^{2+}\ SCN]^{2+}$**

# 5. Charge Transfer Transition

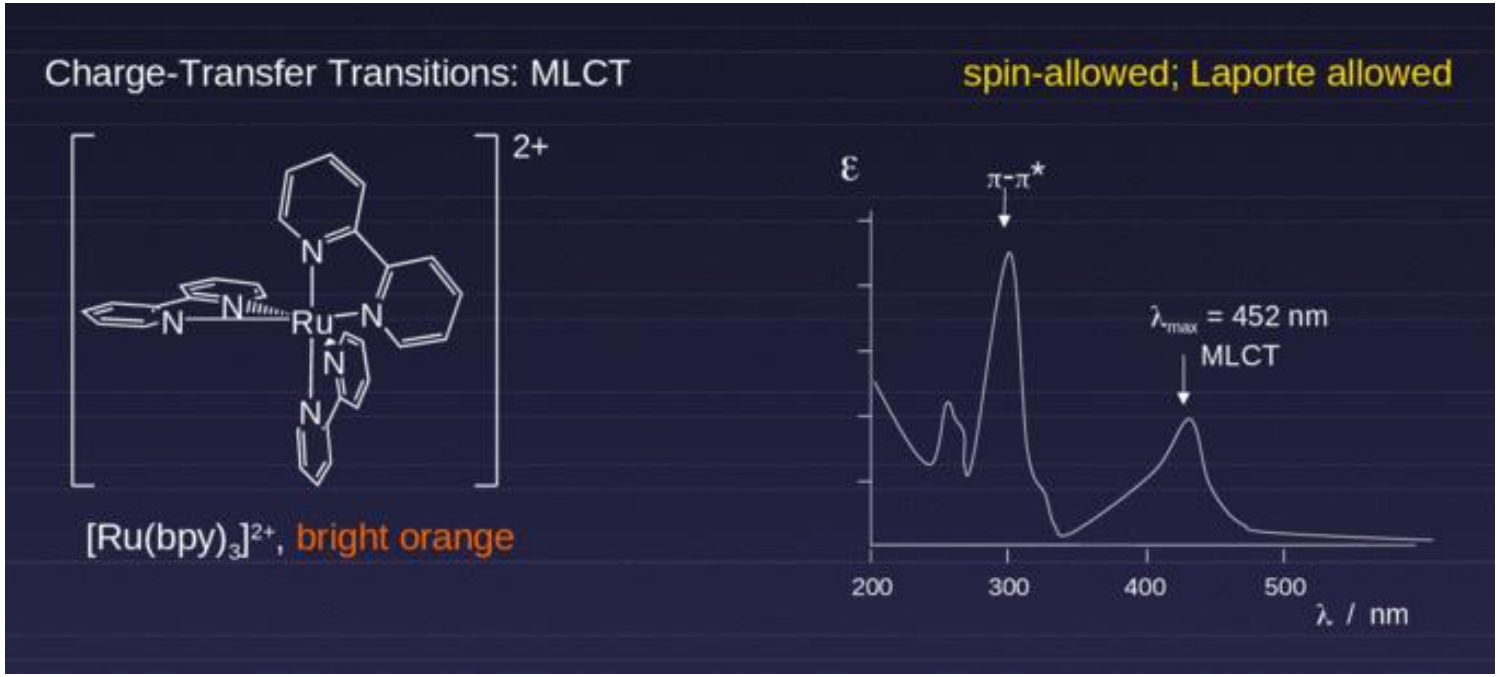
Inorganic compounds

**Features:**

▲ Broad bands

▲ Bands at long wavelengths

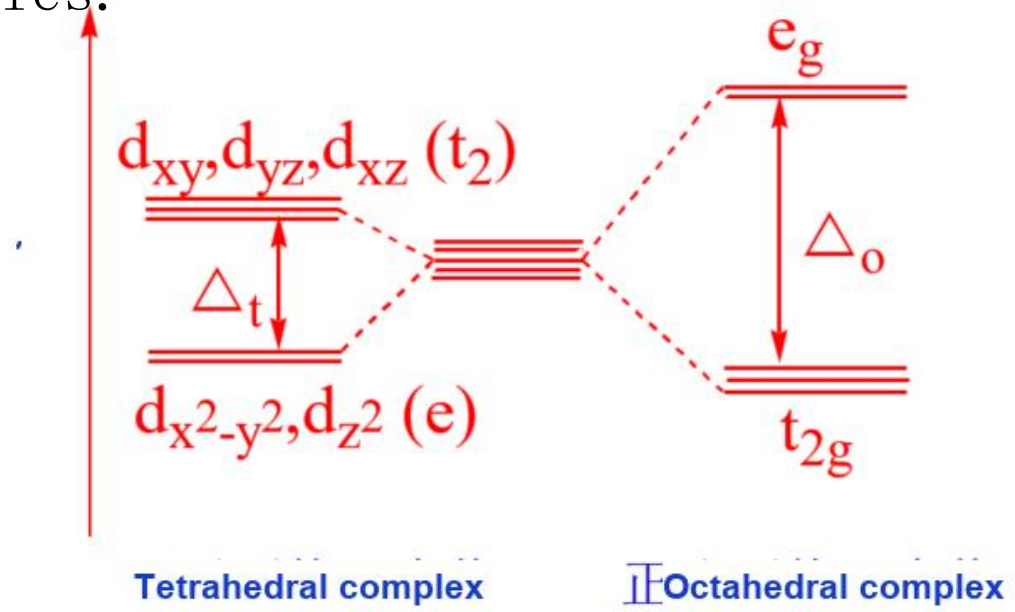
▲ εmax>$10^4$




- MLCT: metal→ligand; LMCT: ligand→metal charge transfer
- LLCT: ligand→ligand; MMCT: metal→metal charge transfer
- Example: MLCT band of $[Ru(bpy)_3]^{2+}$

# 6. Ligand Field Transition

Ligand field transitions include d -d and f -f transitions. Transition metals of periods 4 and 5 have 3d and 4d orbitals; lanthanides and actinides have 4f and 5f orbitals. In the presence of ligands, the five degenerate d orbitals and seven degenerate f orbitals split into groups of different energies.



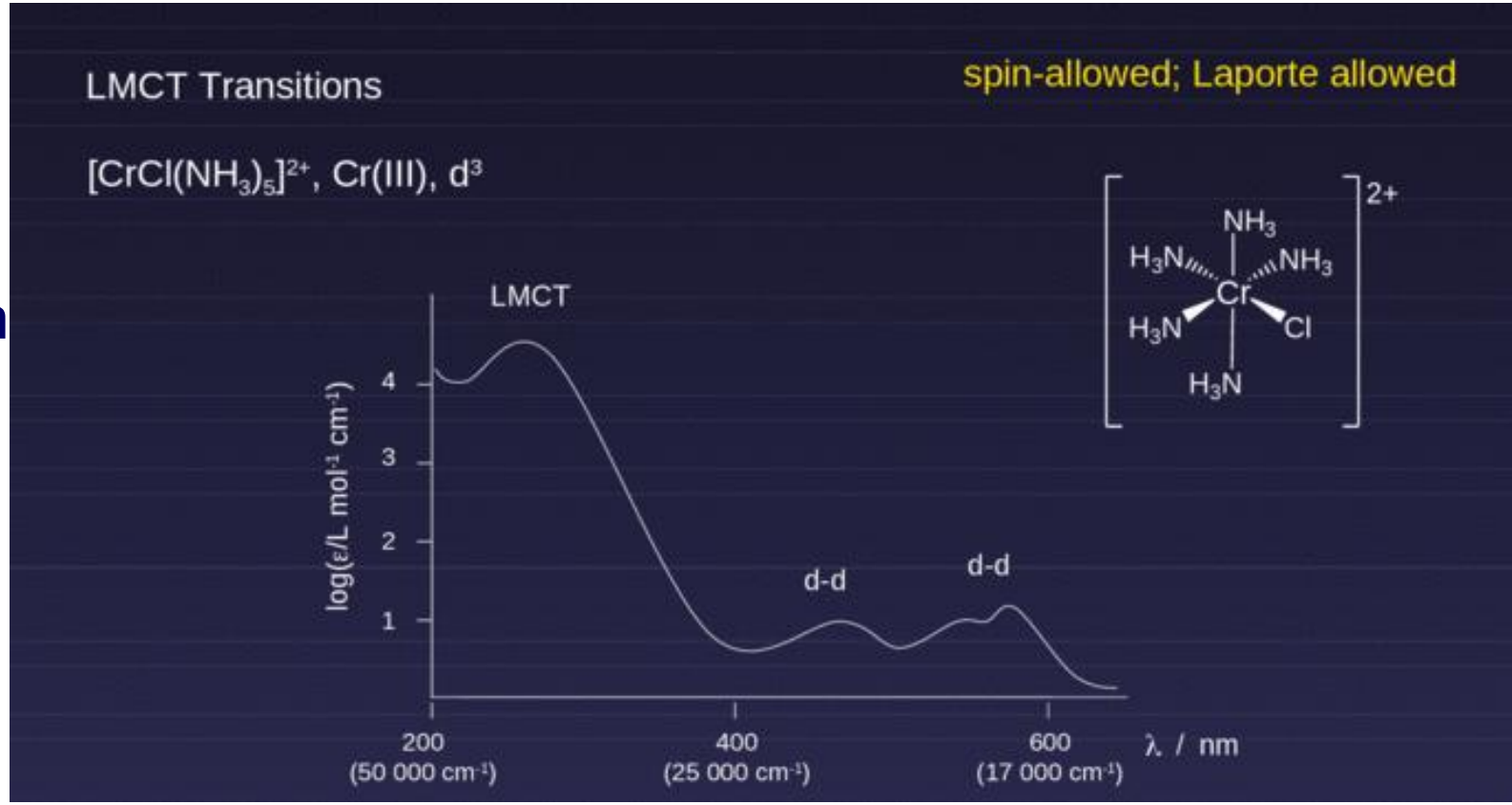
# 6. Ligand Field Transition

## Features:

▲ Broad bands

▲ Bands in the visible region

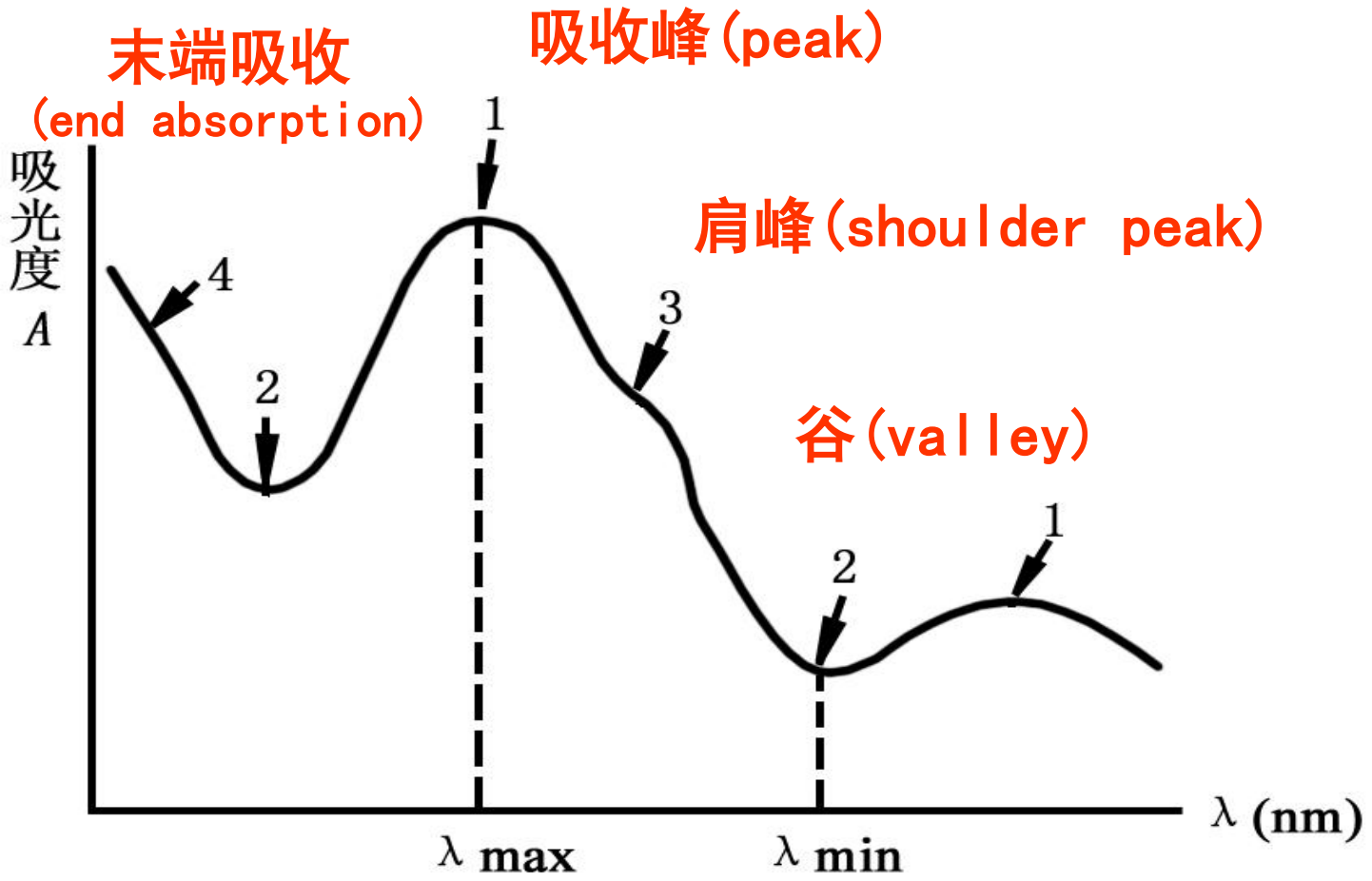
▲ $\varepsilon_{max}<10^2$



# Basic Principles of UV-Vis Spectrophotometry

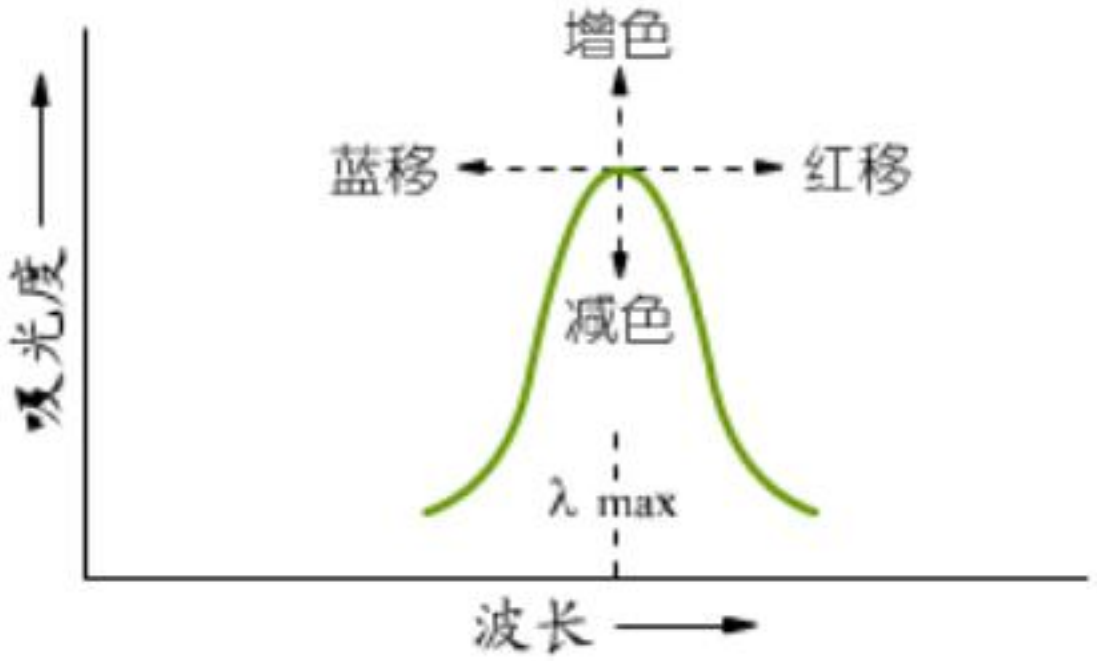
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# Characteristics of Absorption Spectra



# Red Shift and Blue Shift

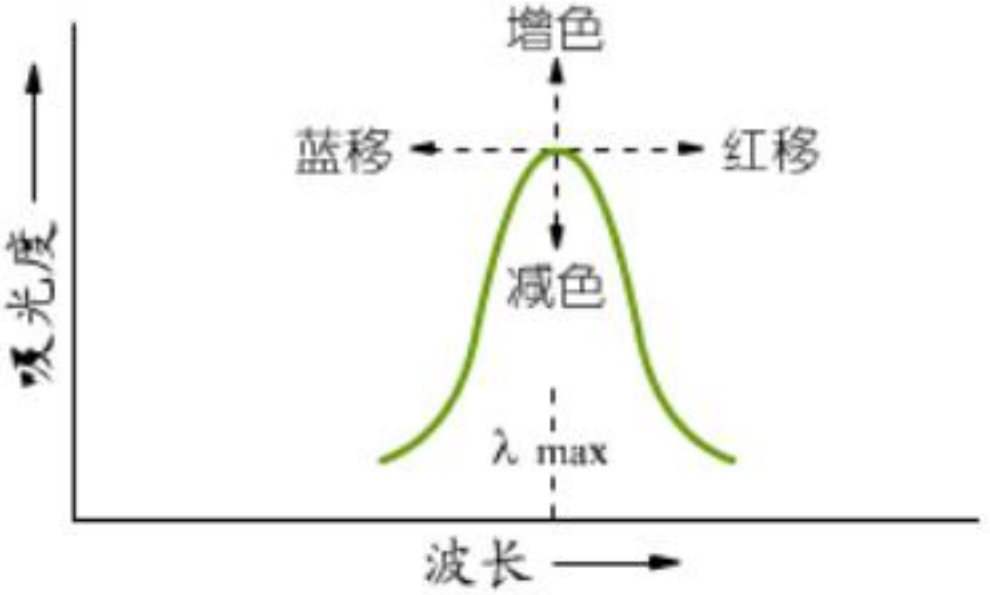
- **Red shift** (bathochromic shift): structural changes such as conjugation, introduction of auxochromes or solvent changes move the absorption peak to longer wavelength.
- **Blue (violet) shift** (hypsochromic shift): structural changes or solvent effects move the peak to shorter wavelength.



# Hyperchromic / Hypochromic Effects; Strong and Weak

- Hyperchromic effects:

Structural changes or other factors increasing absorption intensity: hyperchromic effect; decreasing it: hypochromic effect.




- Strong and weak bands:
- Peaks with $\varepsilon_{max} > 10^4$ are strong bands; $\varepsilon_{max} < 10^2$ are weak bands

# Chromophores

- Definition: groups that absorb in the UV-Vis region

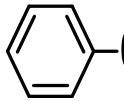
| Chromophore | Solvent | $\lambda$/nm | $\varepsilon_{max}$ | Transition |
|---|---|---|---|---|
| Alkene | n-Heptane | 177 | 13000 | $\pi \rightarrow \pi^*$ |
| Alkyne | n-Heptane | 178 | 10000 | $\pi \rightarrow \pi^*$ |
| Carbonyl | Ethanol | 204 | 41 | $n \rightarrow \pi^*$ |
| Amide | Water | 214 | 60 | $n \rightarrow \pi^*$ |
| Carbonyl | n-Hexane | 186, 280 | 1000 | $n \rightarrow \sigma^*$, $n \rightarrow \pi^*$ |
| Azo | Ethanol | 339，665 | 150000 | $n \rightarrow \pi^*$, |
| Nitro | Isooctane | 280 | 22 | $n \rightarrow \pi^*$ |
| Nitroso | Ether | 300，665 | 100 | $n \rightarrow \pi^*$ |
| Nitrate ester | Dioxane | 270 | 12 | $n \rightarrow \pi^*$ |

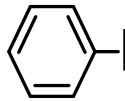
# Auxochromes

- Definition:

Saturated groups with heteroatom non-bonding electrons do not absorb above 200 nm themselves, but with chromophores in the same molecule they shift the peak and increase intensity.


B带 λmax 255nm(ε 230)


–OH B带 λmax 270nm(ε 1450)


–$NH_2$ B带 λmax 280nm(ε 1430)

–Cl, –OH, –$NH_2$ and other auxochromes red-shift the B band and increase its intensity

# Basic Principles of UV-Vis Spectrophotometry

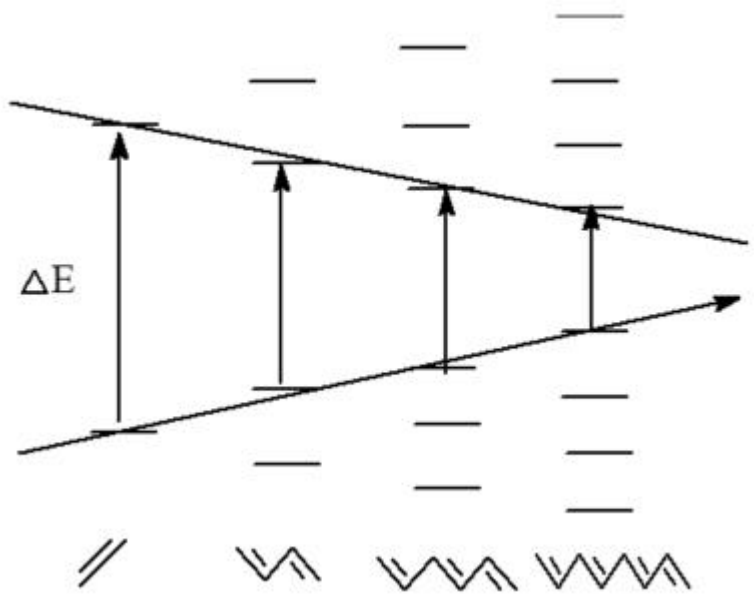
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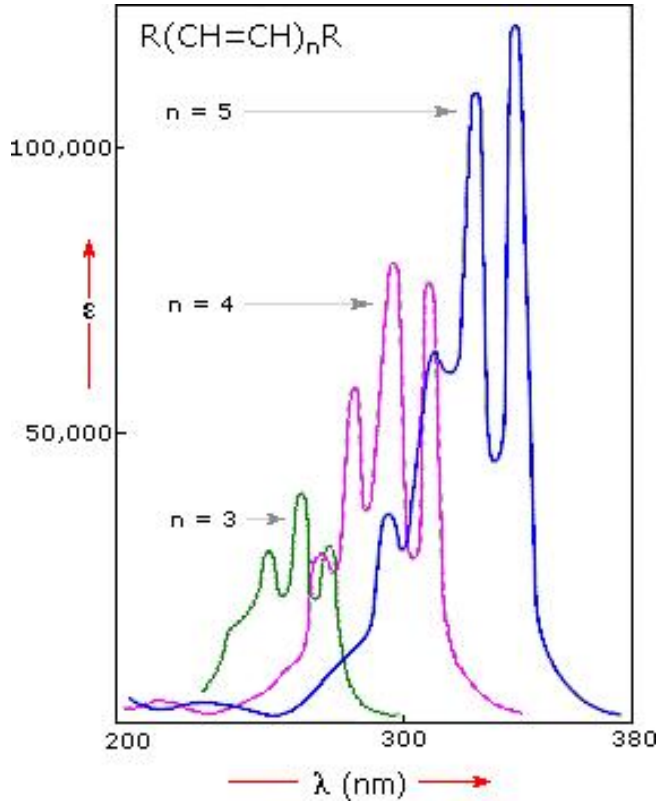
# R Band

- **R band**: caused by n→π* transitions; characteristic of heteroatom unsaturated chromophores such as >C=O, -NO, -$NO_2$, -N=N-. Wavelength ~300 nm; weak absorption, ε<100.

# K Band

- **K band**: produced by π→π* transitions of conjugated double bonds. Peaks at ~210–250 nm, ε>$10^4$; a strong band.



# B Band:

## B Band:

**Caused by π→π* transition and ring vibration;**

- $\varepsilon_{max}$ = 200; $\lambda_{max}$ = 254 nm; broad band with fine structure;
- In polar solvents or with substituents on the ring, the fine structure vanishes and a red shift occurs.

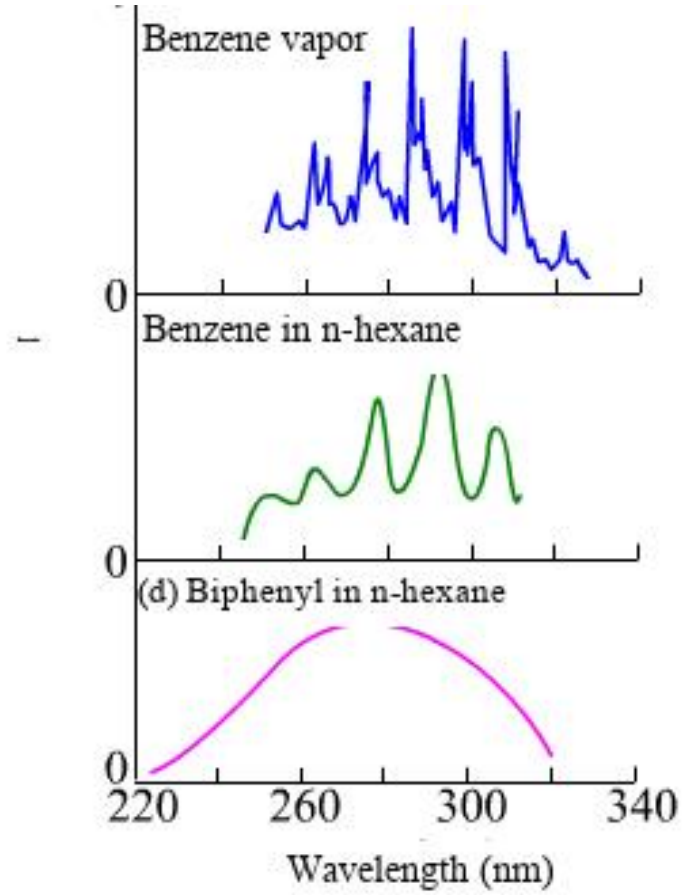



Fig. 4.6 Effect of physical state on the vibrational fine structure of absorption spectra

# E Band

**E band: produced by π→π* transitions of cyclic conjugated systems**

- $E_1$: 180 nm, εmax>$10^4$ (complete peak often unobservable)
- $E_2$: 200 nm, εmax=7000, strong absorption
- When the ring bears a conjugated chromophore, the E band (K band) red-shifts

# Summary of Absorption Band Types

| Band | Character-istics | λ(nm) | Intensity (εmax) | Transition | Absorbing groups |
|---|---|---|---|---|---|
| R band | Longer λ Weak | ~300 | $\varepsilon_{max}<100$ (weak band) | $n \rightarrow \pi^*$ | Unsaturated group with heteroatom |
| K band | Strong | ~210~250 | $\varepsilon_{max}>10^4$ (strong band) | Conjugated $\pi \rightarrow \pi^*$ | Conjugated double bond |
| B band | Character-istic absorption of aromatic compounds | ~230~270 barycentre ~256 | ~200 | Aromatic C=C skeleton vibration and $\pi \rightarrow \pi^*$ within the ring | Delocalized π bond of aromatic ring |
| E band: $E_1$ | Strong | ~180 | ~47000 | Aromatic conjugated $\pi \rightarrow \pi^*$ | Aromatic ring |
| E band: $E_2$ | Medium strong | ~200 | ~7000 | | |

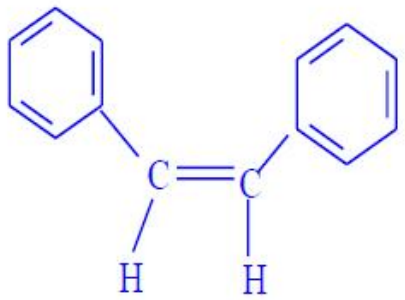
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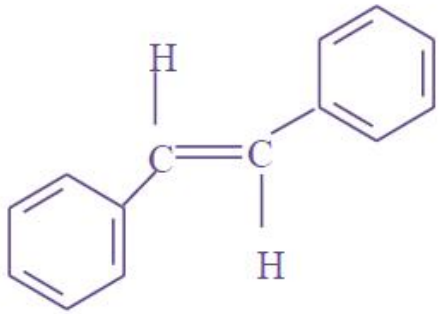
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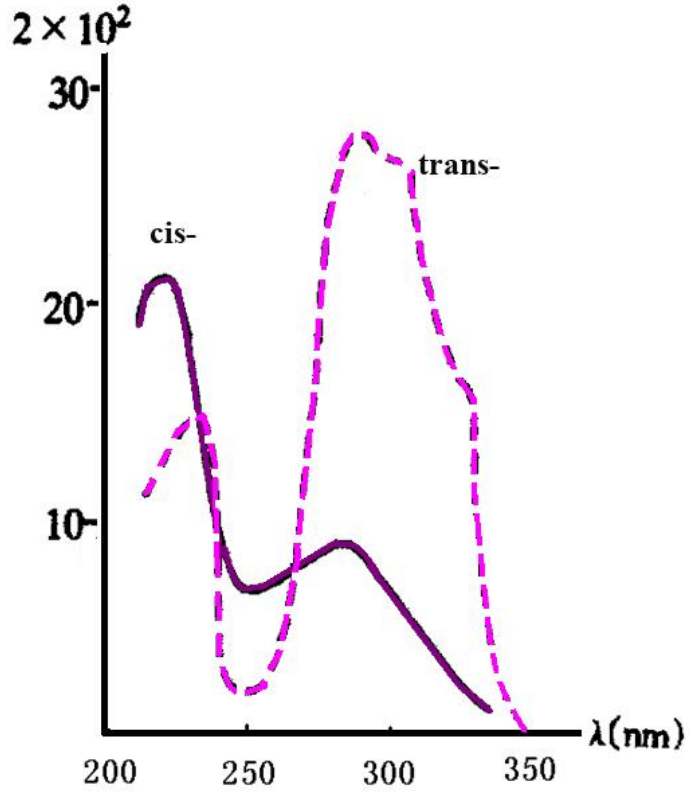
# 1. Steric Effect

- If steric hindrance keeps two chromophores out of the same plane, conjugation is affected.

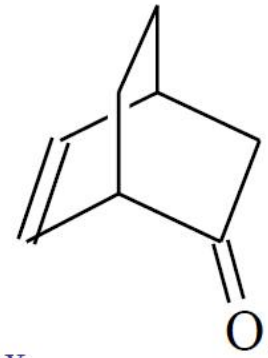
- e.g. stilbene has cis and trans isomers

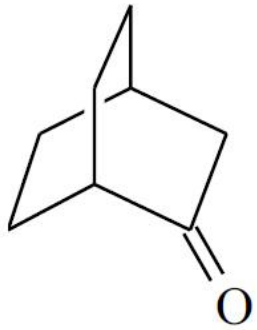


# 2. Transannular Effect




| | | |
|---|---|---|
| $\lambda$ max | 300. 5nm | 280nm |
| $\varepsilon$ max | 292 | ~150 |

- Interaction between non-conjugated groups
- When two non-conjugated chromophores occupy certain spatial positions, especially in cyclic systems, orbital interaction is favored: the transannular effect.

# 3. Solvent Effect

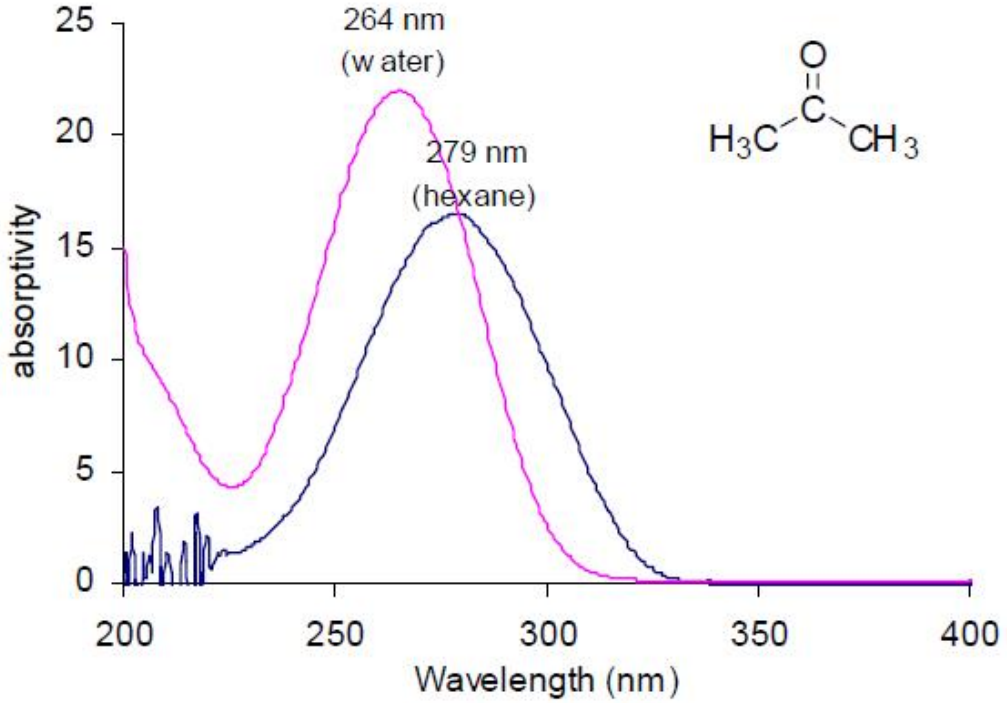



***Figure 5*** *: UV-spectra of acetone in hexane and in water*

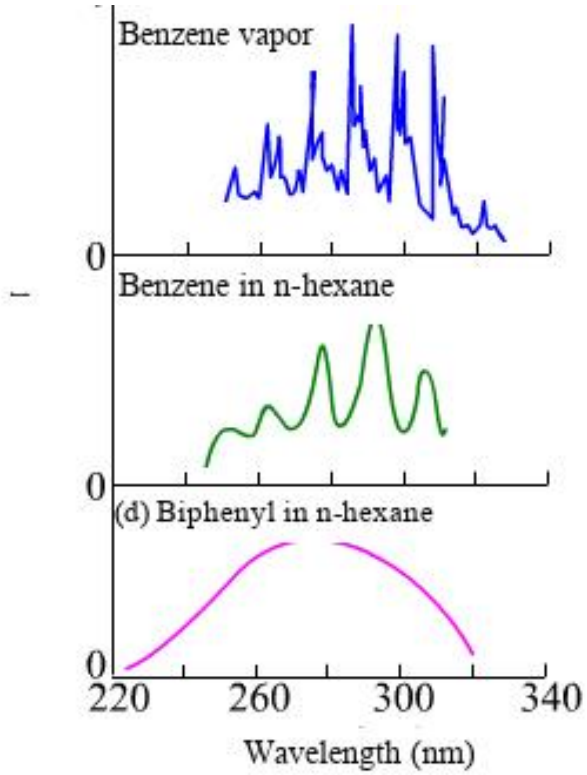
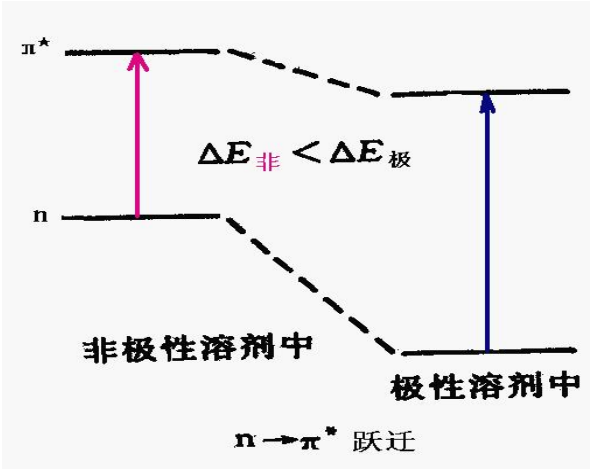



Fig. 4.6 Effect of physical state on the vibrational fine structure of absorption spectra

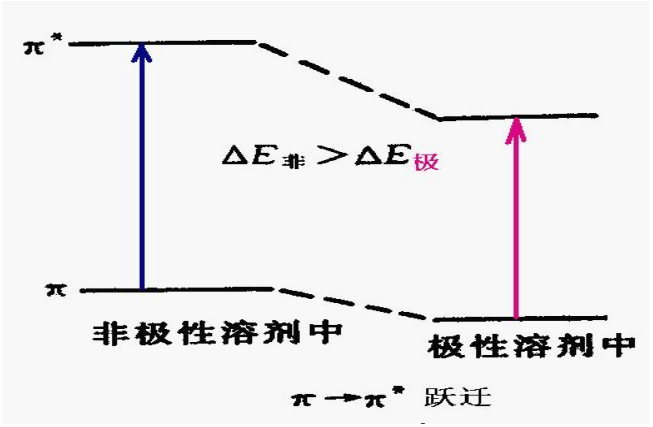
Besides peak position, the solvent also affects intensity and spectral shape.

# Explanation of Solvent Effects

Generally π→π* bands red-shift while n→π* bands blue-shift




The ground state is more polar than the excited state and forms stronger H-bonds with polar solvent, raising the transition energy: blue shift.




The excited state is more polar; with increasing solvent polarity it forms stronger H-bonds and its energy drops more: red shift.

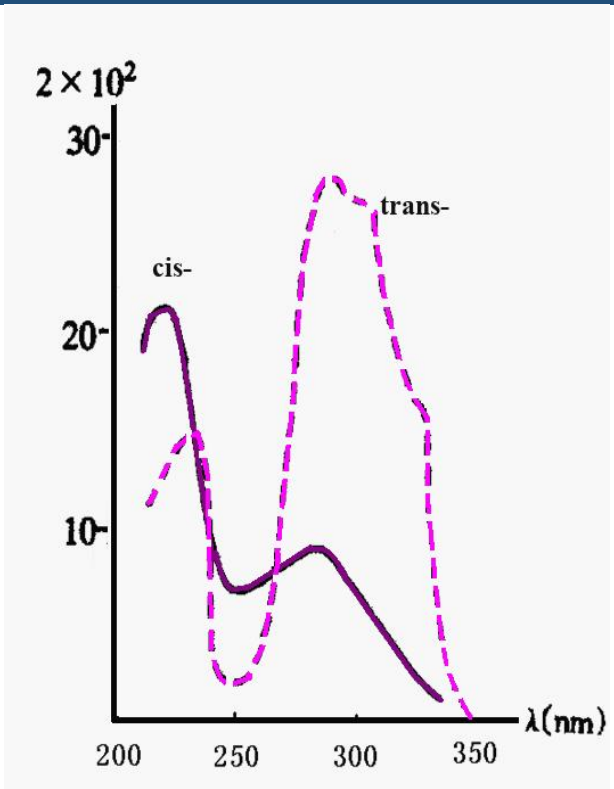
# Rules for Solvent Selection:

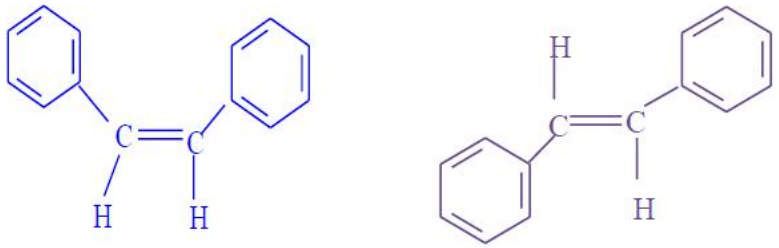
- (1) Different solvents may give different peak shapes and λmax for the same substance; always state the solvent when recording a spectrum
- (2) It should dissolve the analyte well and give a chemically and photochemically stable solution;
- (3) Prefer solvents of low polarity;
- (4) No obvious solvent absorption in the sample region; cutoff < λmax

# Cutoff Wavelengths of Common Solvents

| 溶 剂 | 极限波长 nm | 溶 剂 | 极限波长 nm | 溶 剂 | 极限波长 nm |
|---|---|---|---|---|---|
| 乙醚 | 210 | 96%硫酸 | 210 | 氯仿 | 245 |
| 环己烷 | 200 | 乙醇 | 215 | 四氯化碳 | 260 |
| 正丁醇 | 210 | 正己烷 | 220 | 苯 | 260 |
| 水 | 200 | | | 丙酮 | 330 |

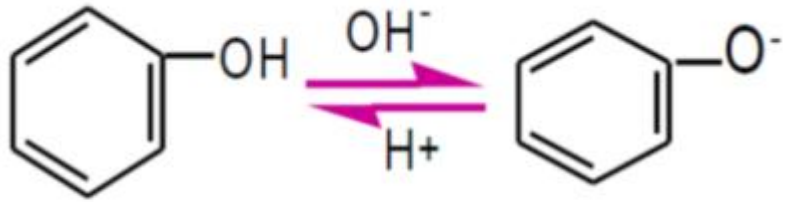
Question: which solvent should be chosen?



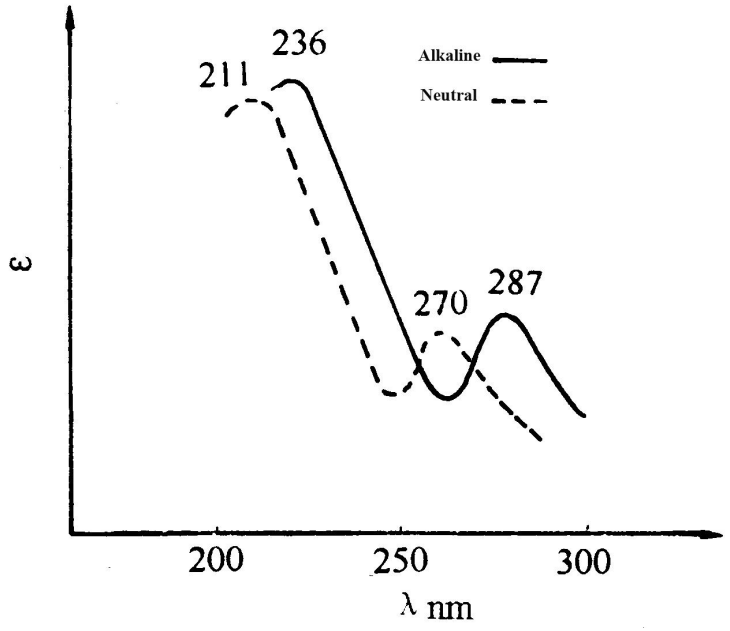
# 4. Effect of pH

When the analyte has acidic or basic groups, solvent pH commonly affects the UV spectrum

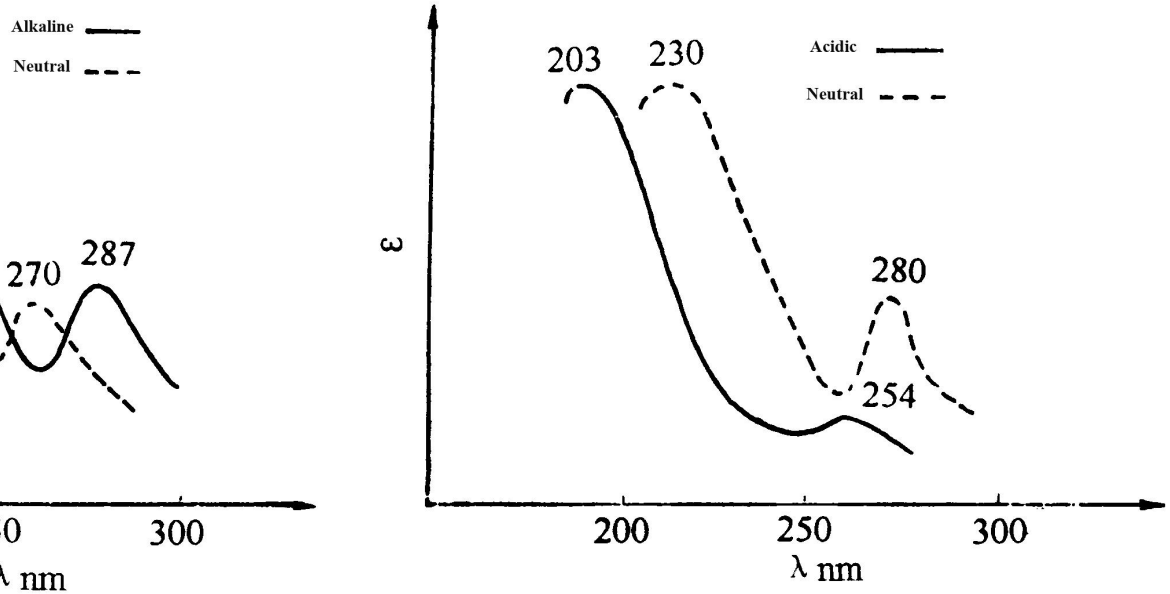



$\lambda_{max}$210.5nm，270nm　　$\lambda_{max}$235nm，287nm

# Examples of pH Effect

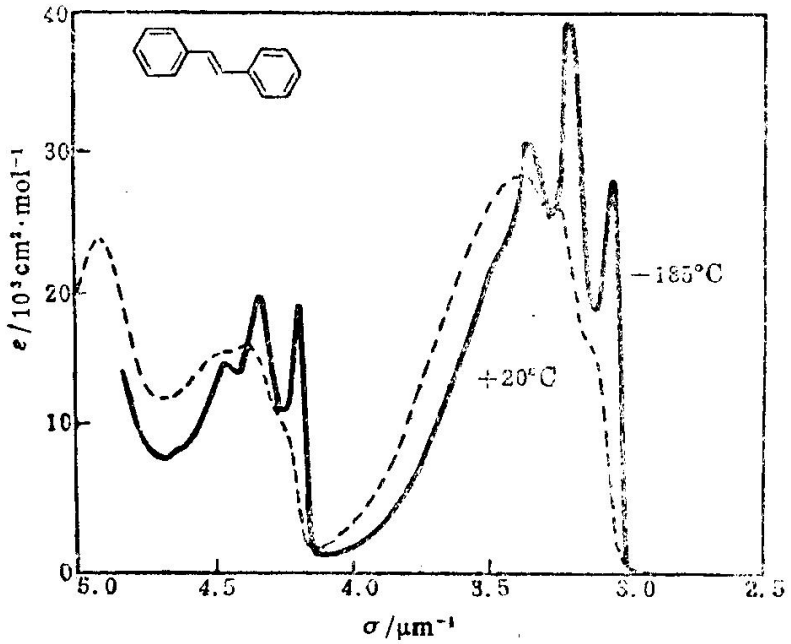



UV spectrum of phenol




UV spectrum of aniline

# 5. Effect of Temperature



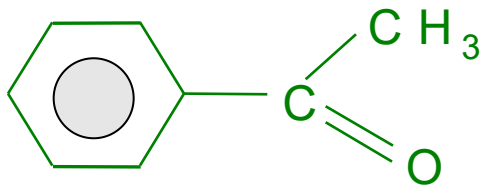

UV spectra of trans-1,2-diphenylethene in isobutane–methylcyclohexane

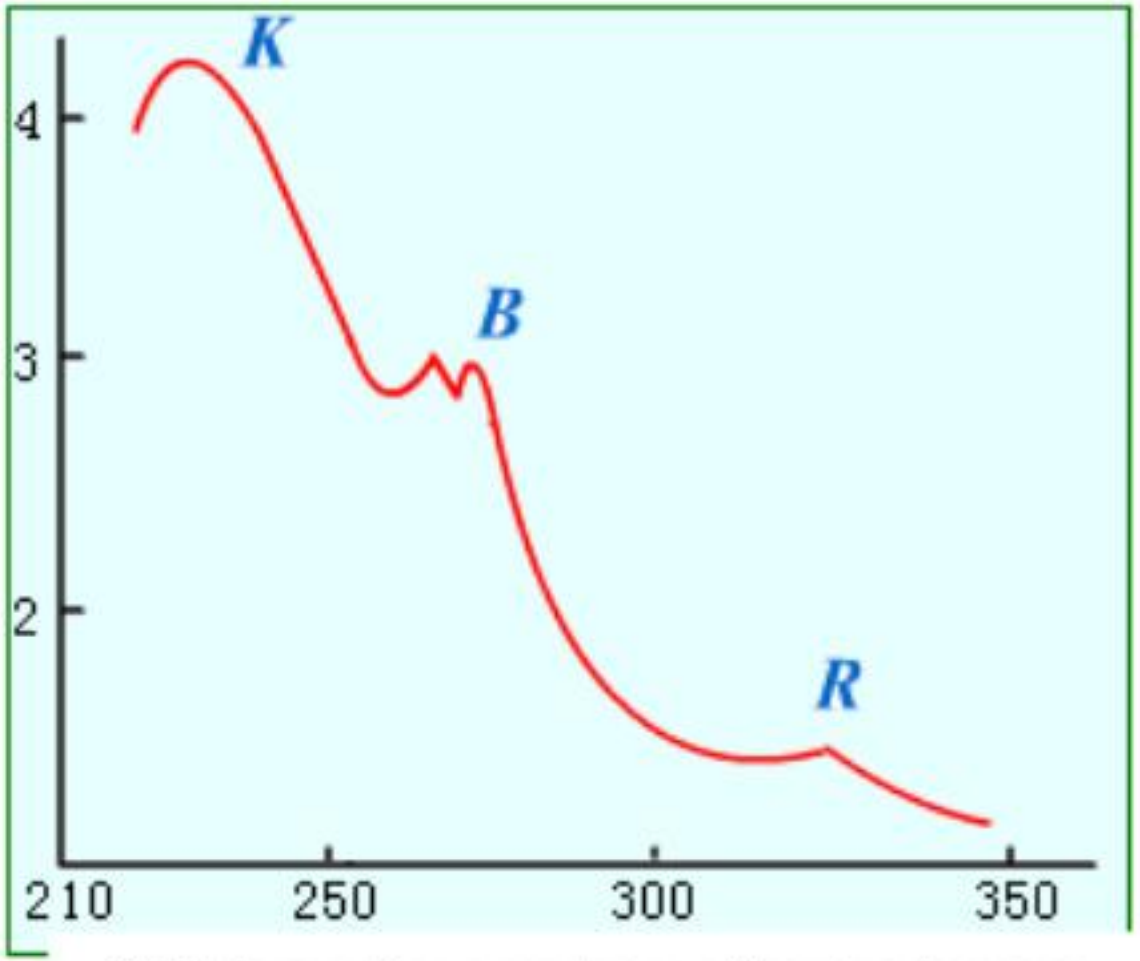
(5:1, v/v)

---- at room temp. (+20°C) —— at −185°C

- Lower temperature reduces vibrational/rotational effects, revealing fine structure

# Example: Predicting the UV Absorption Bands of a Compound




UV absorption spectrum of acetophenone

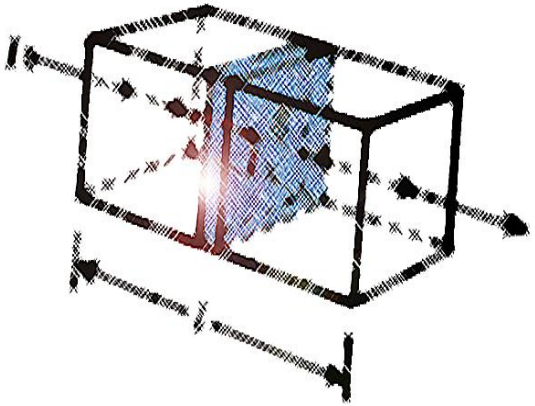
# Basic Principles of UV-Vis Spectrophotometry

- I. Types of Electronic Transitions
- II. Common Terms
- III. Absorption Bands and Molecular Structure
- IV. Factors Affecting Absorption Bands
- **V. Lambert-Beer Law**
- VI. Deviation from the Beer Law

# Lambert-Beer Law

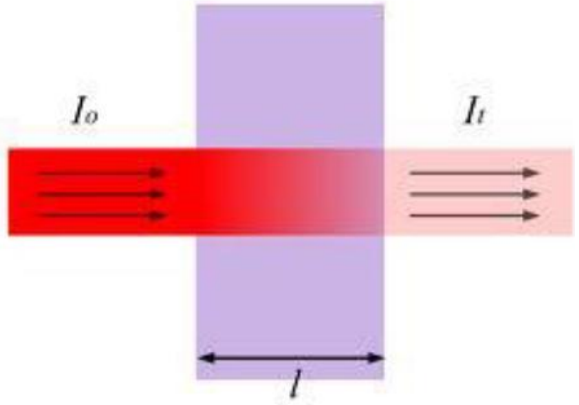
- The fundamental law of absorption photometry: it relates the absorption of monochromatic light to the concentration and thickness of the absorbing substance.

- Conditions
- The solution is dilute (<0.01 mol/L)
- The incident light is monochromatic



# Lambert-Beer Law — Transmittance

公式的推导过程自学




$$-\lg\frac{I}{I_0}=ECL=-\lg T$$

$I_0$ 为入射光强度，$I$ 为透射光强度

$T=\frac{I}{I_0}$ 常用百分数表示，$T\%$

$$-\lg T = ECL$$

**透光率$T$（*transmittance*）与浓度$C$或厚度$L$之间为指数函数的关系。**

# Lambert-Beer Law — Absorbance

## Absorbance

$$-\lg \frac{I}{I_0} = ECL = -\lg T$$

$$A = \lg \frac{I_0}{I} = -\lg T = ECL$$

- Absorbance is directly proportional to concentration or thickness

# Lambert-Beer Law — Absorptivity

$$E = \frac{A}{C \cdot l}$$

- The absorbance at unit concentration and unit thickness; it measures the ability to absorb light of a given wavelength.

**1. 摩尔吸光系数**

在一定波长下，C=1mol/L，L=lcm 时的吸光度，用ε或*E* 标记。

**2. 百分吸光系数或称比吸光系数**

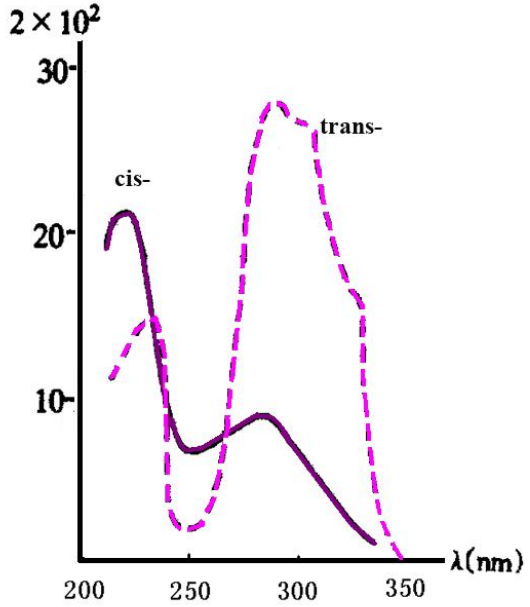
在一定波长下，*C*=1g/100ml，L=lcm 时的吸光度，用 $E_{1cm}^{1\%}$表示。

**3.两种表示方式之间的关系**

$$\varepsilon = \frac{M}{10} \cdot E_{1cm}^{1\%}$$

# Lambert-Beer Law — Absorptivity

- E depends on the substance and wavelength; for a given substance and solvent, E = f(λ)
- E serves for qualitative identification and estimates the sensitivity of quantitative analysis.

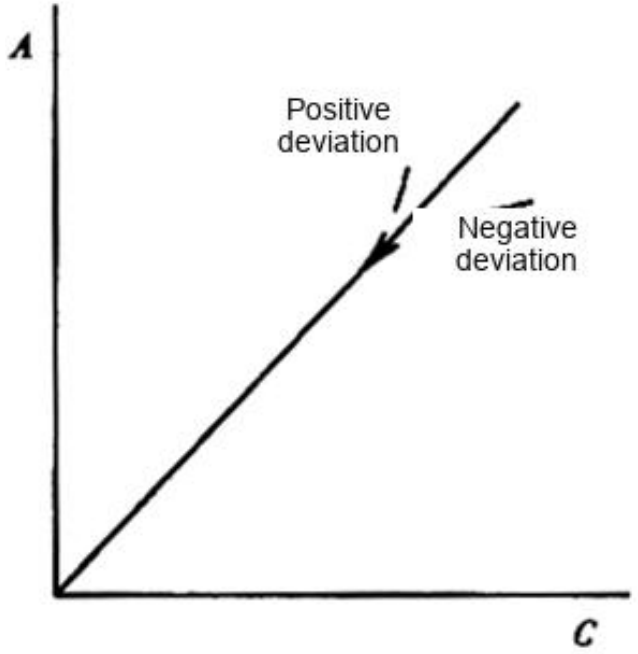


# Basic Principles of UV-Vis Spectrophotometry

- I. Types of Electronic Transitions
- II. Common Terms
- III. Absorption Bands and Molecular Structure
- IV. Factors Affecting Absorption Bands
- V. Lambert-Beer Law
- **VI. Deviation from the Beer Law**

# Factors Causing Deviation from the Beer Law

- Two assumptions were made in deriving the L-B law:
  - the solution is a dilute solution of the absorber (<0.01 mol/L)
  - the incident light is monochromatic




- By the Beer law, A vs C should be a straight line through the origin

# Deviation from the Beer Law — Chemical Factors

- The absorber may associate, dissociate, tautomerize or form complexes, etc.

$$Cr_2O_7^{2-} + H_2O \Leftrightarrow 2H^+ + 2CrO_4^{2-}$$

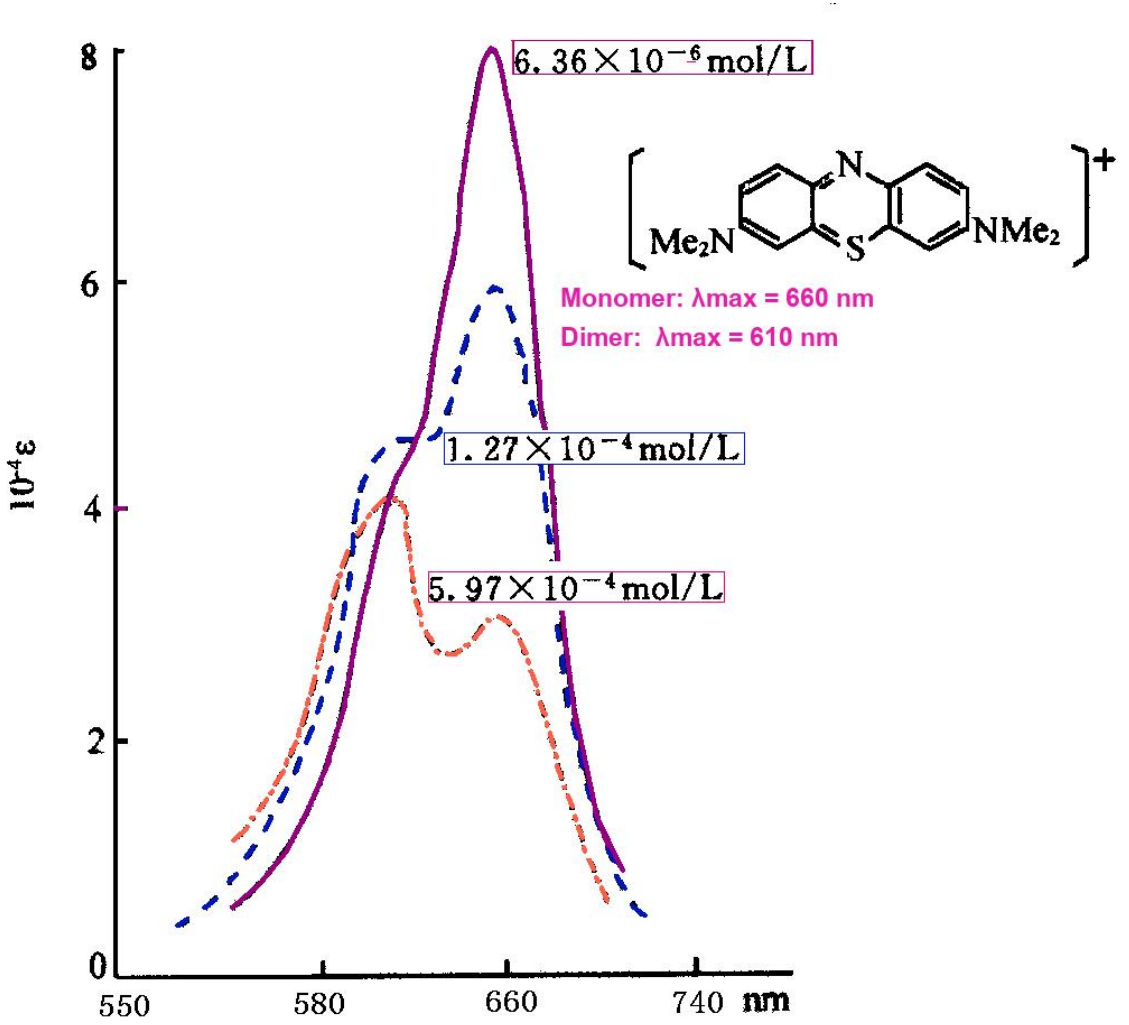
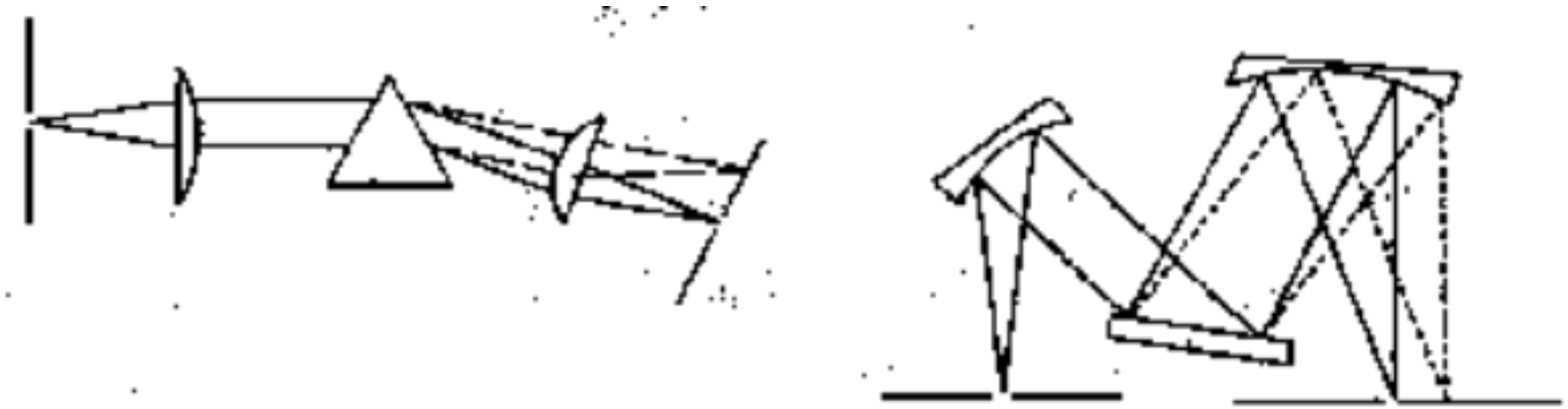



Fig. 11-9 Absorption spectra of methylene blue cation in water

# Deviation from the Beer Law — Optical Factors

## 1. Effect of Non-monochromatic Light

A finite slit width is needed for a detectable response, so the isolated light has a finite bandwidth: light from the monochromator is not truly monochromatic. Its width is set by the slit width and the resolution of the prism or grating.

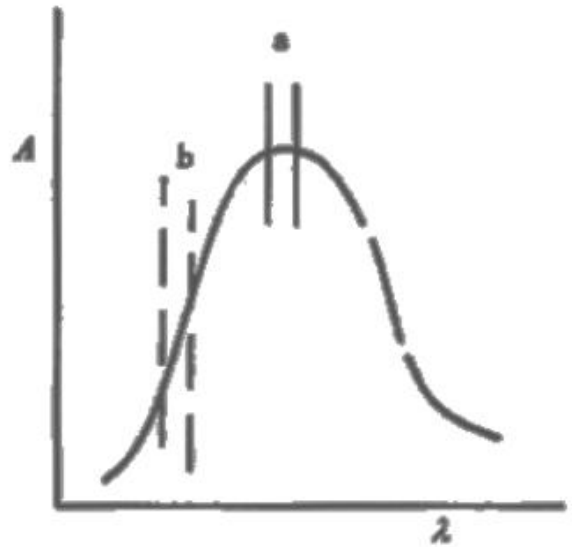


如：复合光（ 入射光强度为 $I_{0_1}+I_{0_2}$），透过光强度为 $I_1+I_2$，所得吸光度值为：

$$A=\lg\frac{I_{0_1}+I_{0_2}}{I_1+I_2}=\lg\frac{I_{0_1}+I_{0_2}}{I_{0_1}10^{-E_1CL}+I_{0_2}10^{-E_2CL}}=E_1cL-\lg\frac{I_{0_1}+I_{02}\cdot10^{(E_1-E_2)cL}}{I_{0_1}+I_{02}}$$

只有当$E_1=E_2$时，$A$与$C$成线性。

（$E_2-E_1$）的差值↑，$A$与$C$偏差线性↑。

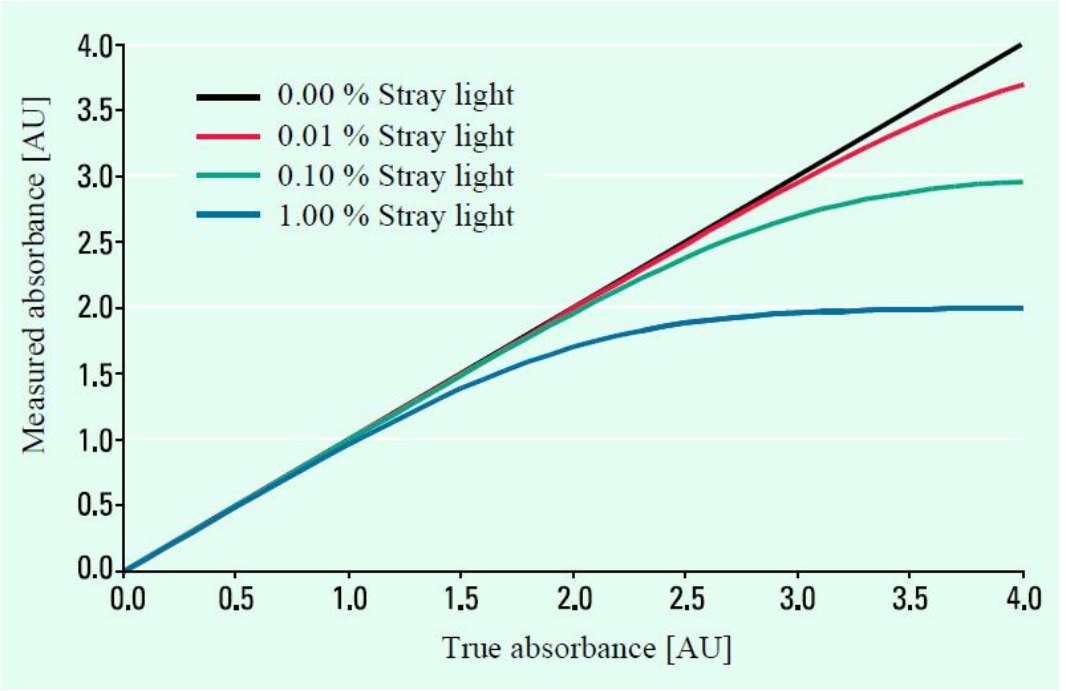



- 对策：
- 选择较纯单色光（$\Delta\lambda\downarrow$，单色性↑）
- 选$\lambda_{max}$作为测定波长（$\Delta E\downarrow$，$S\uparrow$且成线性）

# Deviation from the Beer Law — Optical Factors

## 2. Effect of Stray Light

• Stray light: light from the monochromator outside the incident bandwidth, i.e. light far from the selected wavelength.




• Sources: instrument defects; contaminated optics.

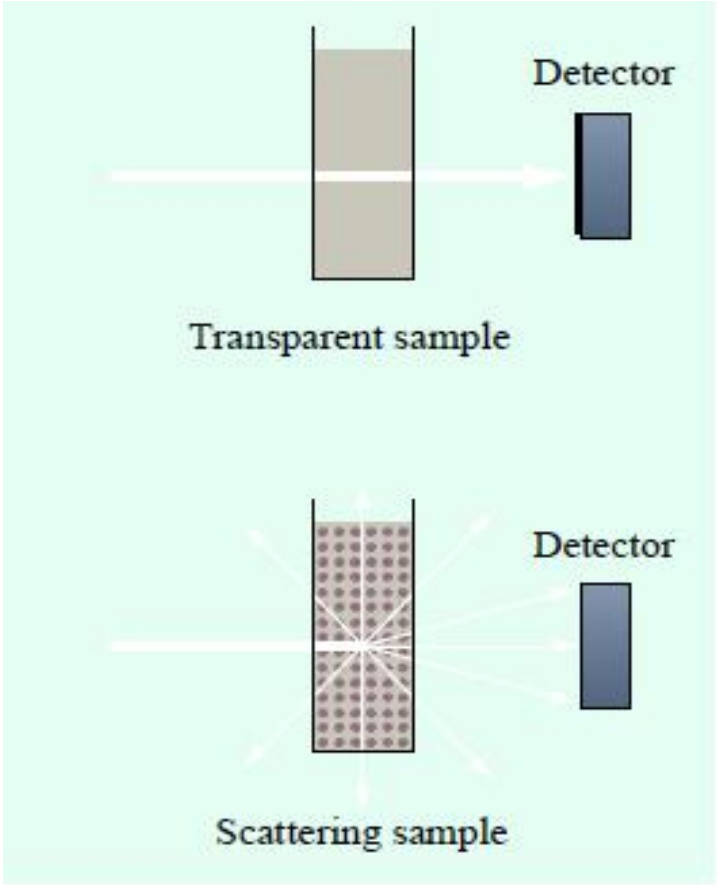
• consequence： Stray light distorts the spectrum and changes A.

• Countermeasure: avoid end-absorption wavelengths

# Deviation from the Beer Law — Optical Factors

## 3. Effects of Reflected and Scattered Light

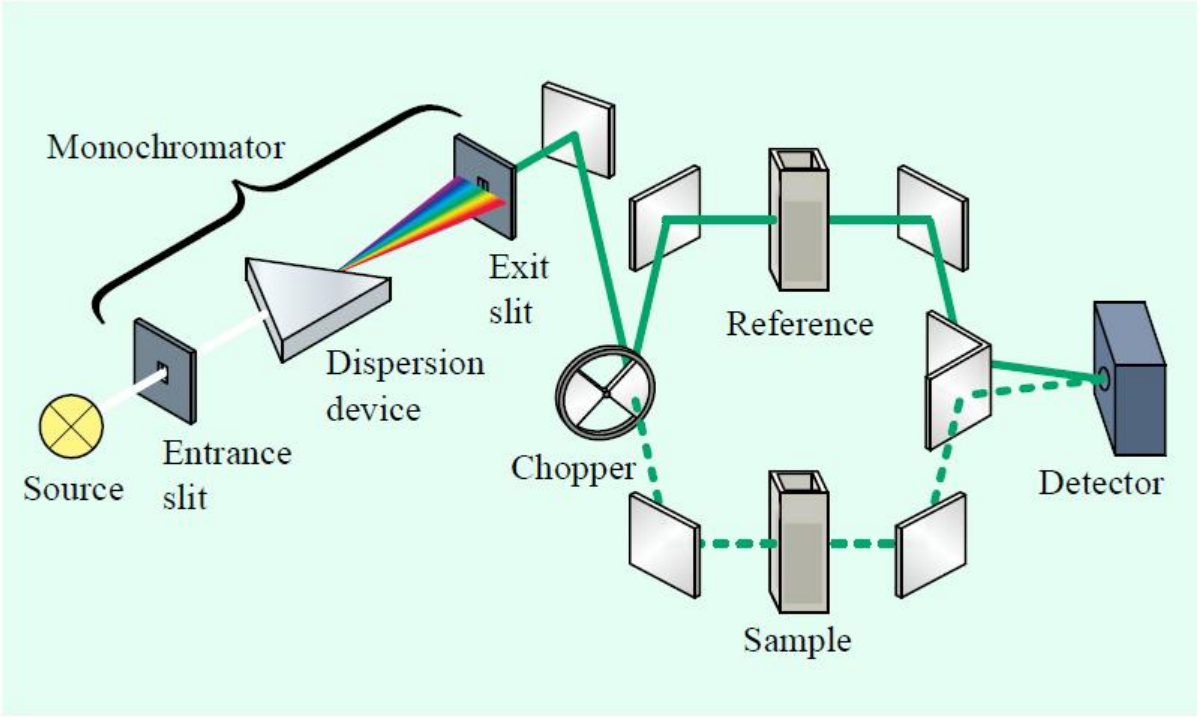
- **Absorbing particles scatter the incident light, and reflections occur at cuvette interfaces; all within the incident bandwidth, directly affecting T.**
- **Scattering and reflection decrease T, increase A, and distort the spectrum.**



# Deviation from the Beer Law — Optical Factors

## 3. Effects of Reflected and Scattered Light

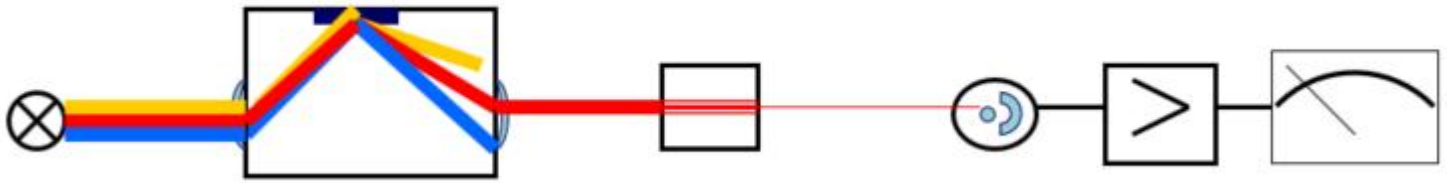
- **Usually eliminated by blank correction**



# Deviation from the Beer Law — Optical Factors

## 4. Effect of Non-parallel Light

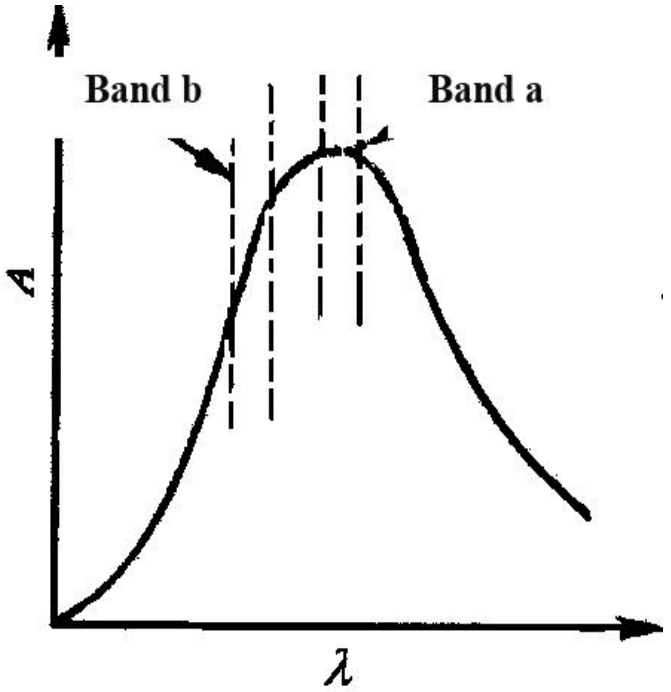
➢Oblique light has a longer path through the cuvette: A ↑ and the spectrum distorts. This is a main reason absorptivity differs between instruments.



# Deviation from the Beer Law — Measurement of absorbance

## 1. Selection of Absorption Wavelength

Usually λmax of the substance is chosen as the analytical wavelength.




**Higher sensitivity, little change in absorptivity, and smaller deviation from the Beer law.**

# Deviation from the Beer Law — Measurement of absorbance

## 2. Selection of Solvent

Use only regions where the solvent absorbs weakly, to avoid interference.

| Solvent | Cutoff λ (nm) | Solvent | Cutoff λ (nm) | Solvent | Cutoff λ (nm) |
|---|---|---|---|---|---|
| Ether | 210 | 96% $H_2SO_4$ | 210 | Chloroform | 245 |
| Cyclohexane | 200 | Ethanol | 215 | $CCl_4$ | 260 |
| n-Butanol | 210 | n-Hexane | 220 | Benzene | 260 |
| Water | 200 | | | Acetone | 330 |

# Deviation from the Beer Law — Measurement of absorbance

## 3. Selection of Concentration (Transmittance Error)

Measurement errors arise from unstable light source, accidental changes in experimental conditions, and inaccurate readings.

$$A = -\lg T = E \cdot C \cdot l \qquad C = \frac{A}{E \cdot l} = \frac{1}{E \cdot l \cdot \lg T}$$

Relative error of concentration $$\frac{\Delta C}{C} = \frac{0.434 \Delta T}{T \lg T}$$

The above equation shows: the relative error ΔC/C of the measured concentration depends on T of the solution and ΔT of the instrument.

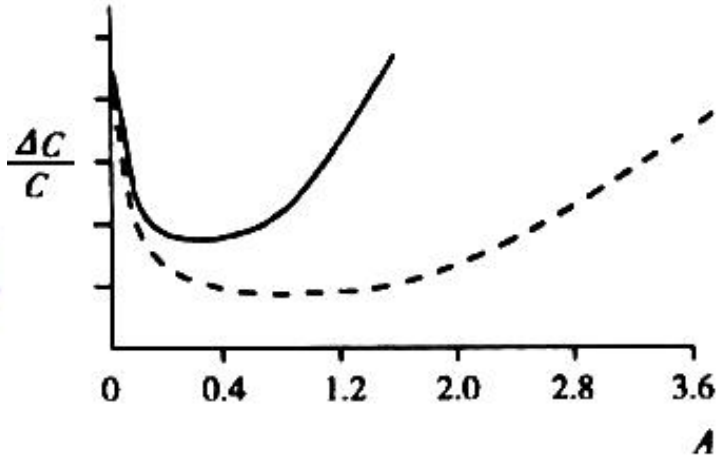
- Factors affecting ΔT: instrument noise ( dark noise ; signal noise )

# Deviation from the Beer Law — Measurement of absorbance

## 3. Selection of Concentration (Transmittance Error)

(1) Dark noise:Related to the detector and amplifier circuit, independent of the light signal.

$$\frac{\Delta C}{C} = \frac{0.434\Delta T}{T \lg T}$$




Error curves for dark noise (—) and signal noise (···)

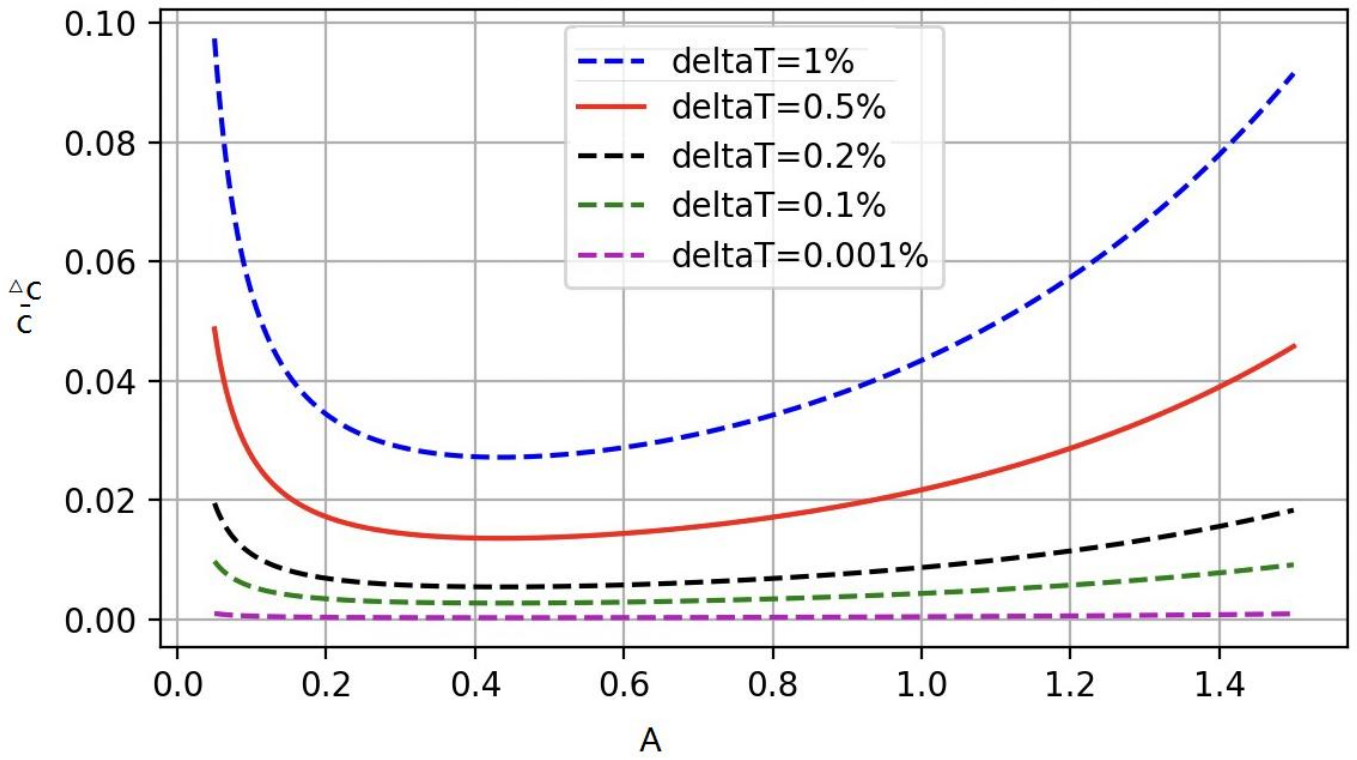
**e.g.**

**Assuming ΔT = 0.5%, substitute into the equation to get the relative error at different T or A:**

- **At T = 0.368 (A = 0.4340), △C/C is minimal: smallest measurement error.**
- **For A = 0.2–0.7 (T = 70%–20%), the error is ~1.4–1.6%: the optimal measuring range.**

P170, Exercise 3

# 3. Selection of Concentration (Transmittance Error)



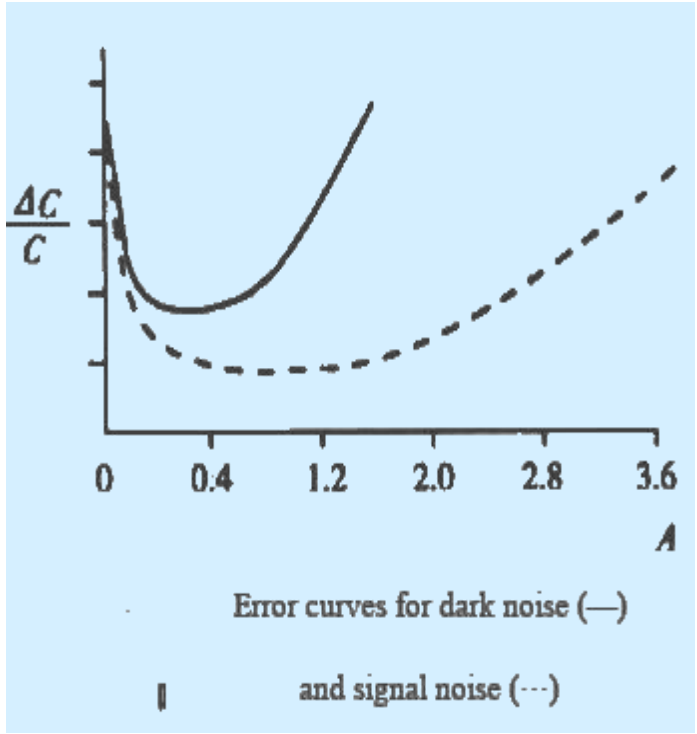

P170, Exercise 3

- Note: today's best UV-Vis spectrophotometers have $\Delta T = 0.01\%$; at 0.02 Abs the error can reach 1%

# Deviation from the Beer Law — Measurement of absorbanc

## 3. Selection of Concentration (Transmittance Error)

(2) Signal noise: related to the light signal




Error curves for dark noise (—)

and signal noise (···)

# 03

# UV-Visible Spectrophotometers

# 04

# UV Absorption Spectra and Structure of Organic Compounds

05

# Qualitative Analysis Methods

# I. Qualitative Identification

**Comparison method:**

**Sample x**

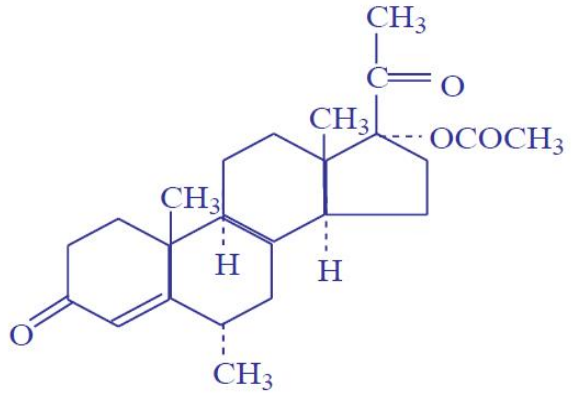
**Standard s**

} **measure spectra in the same solvent under the same conditions.**

# I. Qualitative Identification

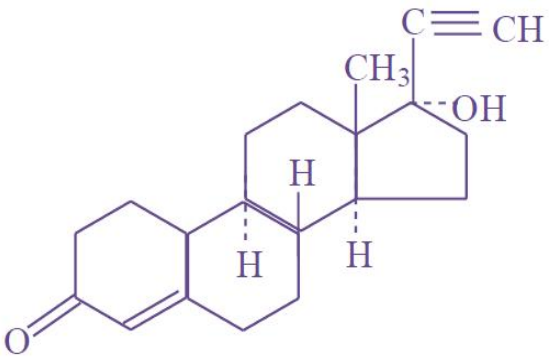
1. Compare spectral characteristic data:

- Position (wavelength) of peak, valley or shoulder
- Absorptivity of the peak: ελmax or E(1%,1cm)λmax




安宫黄体酮
(M=386.53)
$\lambda_{max}$=240±1nm

$E_{1cm}^{1\%}$ =408



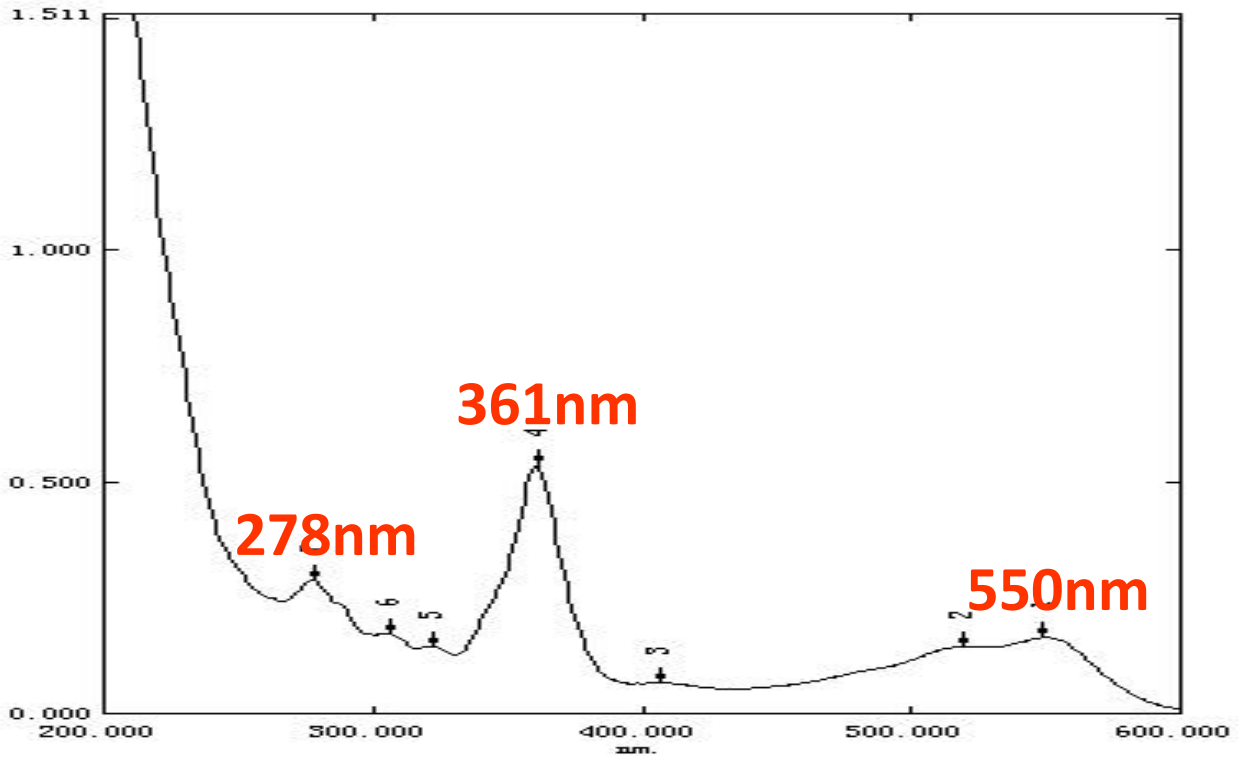

炔诺酮
(M=298.43)
$\lambda_{max}$ =240±1nm

$E_{1cm}^{1\%}$=571

# I. Qualitative Identification

## 2. Compare the Ratio of Absorbances

$$\frac{A_1}{A_2} = \frac{E_1 CL}{E_2 CL} = \frac{E_1}{E_2}$$

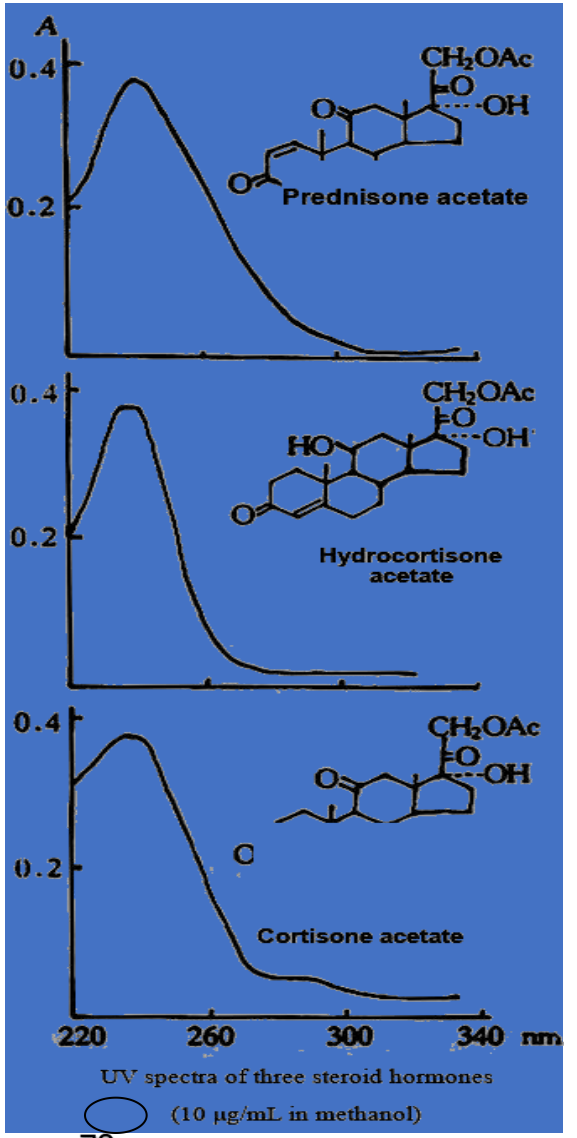



[Identification] The solution from the assay section, measured by UV-Vis (Appendix IV A), shows maxima at 361 nm and 550 nm; A(361)/A(550) should be 3.15–3.45.

# I. Qualitative Identification

## 3. Compare the Consistency of Absorption Spectra

Prepare sample and standard solutions of the same concentration, record spectra under the same conditions and compare. Identical compounds give identical spectra; literature spectra may also be used.



UV spectra of three steroid hormones (10 μg/mL in methanol)

# II. Purity Check

## 1. Impurity Check

### (1) Peaks do not overlap

**Main component transparent, impurity absorbs → check the impurity directly.**

**Example: trace benzene in ethanol or cyclohexane absorbs at 256 nm, where the solvents do not absorb.**

### (2) Peaks overlap

Strong main absorption, no/weak impurity absorption → compare with pure standard, E↓;
strong impurity absorption >> main → compare, E↑.

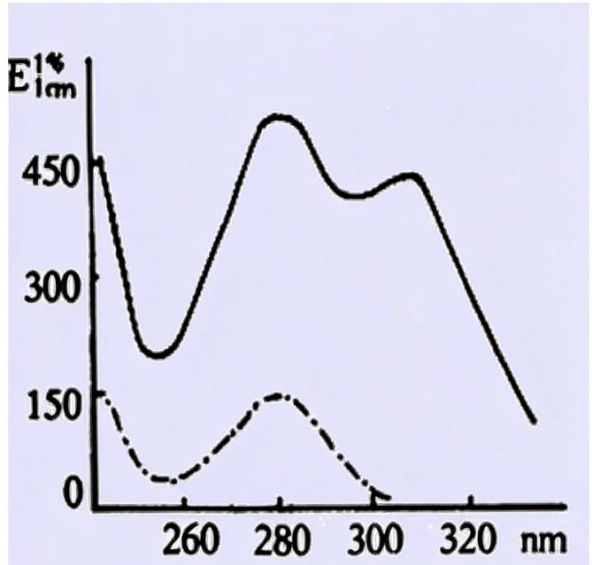
# II. Purity Check

## 2. Limit Test for Impurities

Example: Determination of Adrenalone as a Minor Impurity in Epinephrine

2 mg/mL in 0.05 mol/L HCl, measured at $\lambda_{31}0$ nm

Limit: $A_{310} \leq 0.05$, corresponding to impurity limit ≤ 0.06%




Absorption spectra of epinephrine (solid line) and adrenalone (dashed line)

$$C = \frac{A}{E^{1\%}_{1cm} \cdot l} = \frac{0.05}{435 \times 1}$$

$$= 1.1 \times 10^{-4}\, g/100ml$$

$$= 1.1 \times 10^{-3}\, mg/ml$$

$$\text{Adrenalone } \% = \frac{1.1 \times 10^{-3}}{2} \times 100 = 0.06\%$$

豆包AI生成

# II. Purity Check

- **Sometimes the peak-to-valley absorbance ratio limits impurities**
- For pralidoxime iodide the peak-to-valley ratio is the impurity index: pure $A_{294}/A_{262}$ = 3.39; impurities raise absorption at 262 nm and lower the ratio. A minimum allowable ratio may be specified.

# 06

# Quantitative Analysis Methods

# Overview of Single-Component Quantitative Methods

- Absorptivity method (absolute): A = ECL
- Standard curve (working curve) method
- Comparison method (one-point external standard)
- Colorimetric method

# Single-Component Quantitative Methods

## (1) Absorptivity Method (Absolute Method)

According to Beer's Law: $C = A/EL$

If $L$ and $E$ are known, measure $A$ to calculate $C$.

$$W_{\text{sample (g)}} \xrightarrow{\text{dilution}} C_{\text{sample (g/mL)}} \xrightarrow{E_{\text{known}}} \text{measure } A \rightarrow C_{\text{i}}$$

$$C_{\text{i}(g/100mL)} = \frac{A}{E \cdot l} \quad \text{or} \quad C_{\text{i(mol/L)}} = \frac{A}{\varepsilon \cdot l}$$

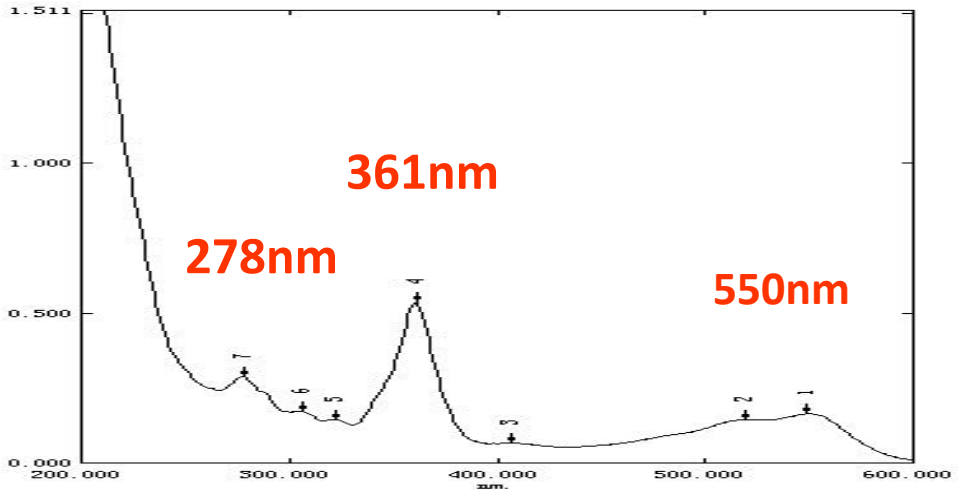
Prerequisite: When using this method, the absorptivity should generally be greater than 100. Calibration and verification of the instrument should be performed.

# Single-Component Quantitative Methods

- **Example: vitamin $B_{12}$ in water has E(1%,1cm) = 207 at 361 nm. With a 1 cm cell, A = 0.414. Find the concentration.**

  - Solution:

$$C = \frac{A}{E \cdot l} = \frac{0.414}{207 \times 1} = 0.00200\,g/100ml = \mathbf{0.0200}\ \mathrm{mg/mL}$$



# Single-Component Quantitative Methods

Example 2: 25.0 mg of vitamin $B_{12}$ is dissolved in water to 1000 mL; in a 1 cm cell at 361 nm, A = 0.507. Find the percent content of $B_{12}$.

- **Solution:**

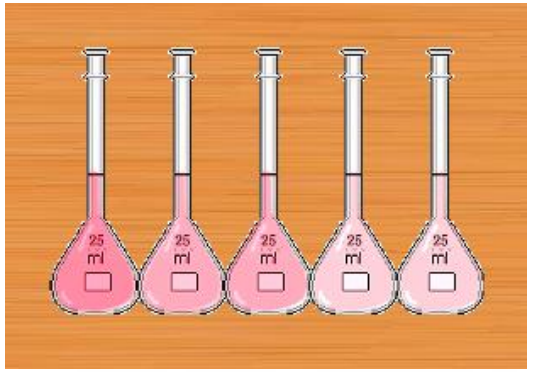
$$C_i = \frac{0.507}{207 \times 1} = 2.45 \times 10^{-3}\,\text{g}/100\text{ml} = 2.45 \times 10^{-5}\,\text{g}/\text{ml}$$

$$C_{smp} = \frac{2.50 \times 10^{-2}}{100} \times \frac{10}{100} = 2.50 \times 10^{-5}\,g/mL$$

$$B_{12}\,\% = \frac{C_i}{C_{smp}} \times 100\% = \frac{2.45 \times 10^{-5}}{2.50 \times 10^{-5}} \times 100 = 98.0$$

# Single-Component Quantitative Methods

## (2) Standard (Calibration) Curve Method



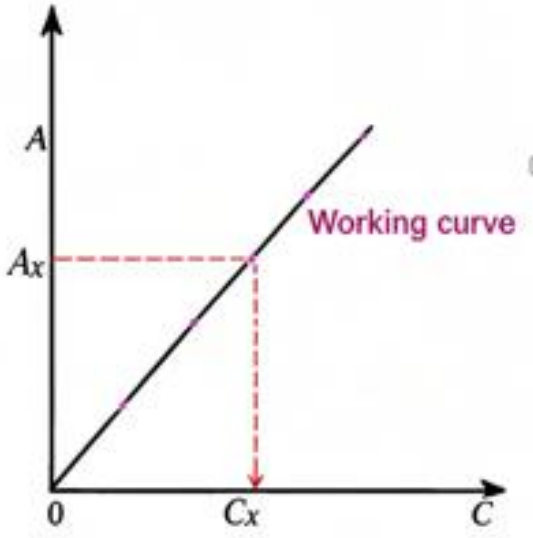

**Core Principle** Under fixed instrument and measurement conditions, absorbance A is proportional to concentration C.

$$A = K \cdot C$$

$$A \propto C$$

**Procedure** Prepare a standard series with known concentrations C. Measure the absorbance A of each standard under fixed conditions. Plot the calibration curve of C versus A.

For sample measurement: Measure the absorbance of the sample under the same conditions. Read the corresponding concentration from the calibration curve.

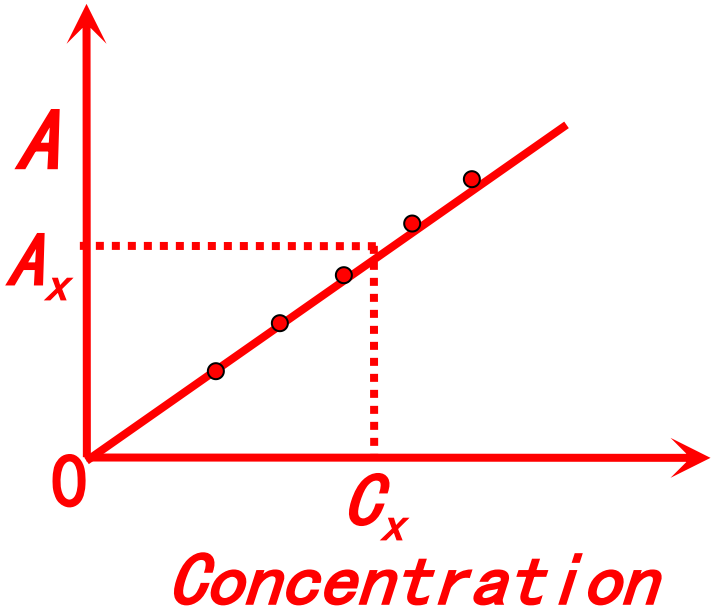


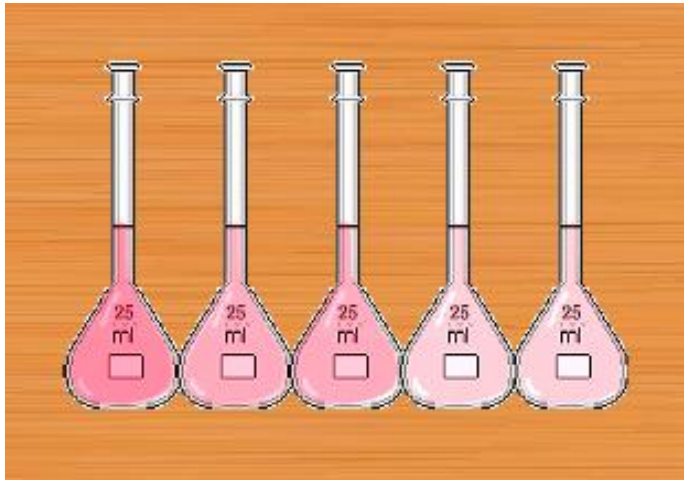

**Key Requirement** Instrument parameters and measurement conditions must remain constant during both standard and sample testing.

# Single-Component Quantitative Methods

Calibration curve

(A=aC+b, r)



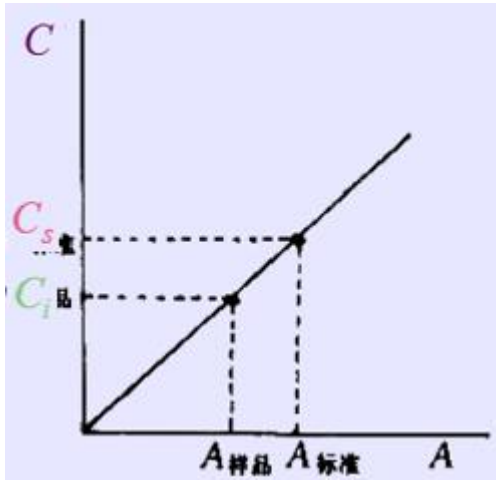

Requirements

1. r > 0.999
2. n = 5-7
3. Sample concentration should lie near the middle of the curve

# Single-Component Quantitative Methods

## 3、Comparison Method (One-Point External Standard)

**Premise** Fixed instrument and measurement conditions. Intercept is 0. Standard and sample concentrations are similar.




**Method**

At fixed λ, measure $A_{unk}$ and $A_{std}$.

$$A_{unk} = C_{unk} \cdot E \cdot l$$

$$A_{std} = C_{std} \cdot E \cdot l$$

$$\frac{A_{unk}}{A_{std}} = \frac{C_{unk}}{C_{std}}$$

$$C_{unk} = C_{std} \times \frac{A_{unk}}{A_{std}}$$

This method requires a reference standard but can partly compensate for variations in wavelength accuracy, slit width, and other measurement conditions; therefore, it is a recommended method.

# Dissociation Constants of Weak Acids/Bases by UV-Vis

Principle: when the acidic/basic group is part of the chromophore, the spectrum changes with pH:

Acid: $HA = H^+ + A^-$,

Low pH: spectrum of HA; high pH: spectrum of $A^-$.

Low pH or high pH $A_{HA} = \varepsilon_{HA} \cdot C$ *or* $A_{A^-} = \varepsilon_{A^-} \cdot C$

at the same wavelength

When pH is between the two:

$$A = \varepsilon_{HA} \cdot [HA] + \varepsilon_{A^-} \cdot [A^-] = \varepsilon_{HA} \cdot \frac{C \cdot [H^+]}{K_a + [H^+]} + \varepsilon_{A^-} \cdot \frac{C \cdot K_a}{K_a + [H^+]}$$

$$\Rightarrow \quad K_a = \left(\frac{A_{HA} - A}{A - A_A}\right) \cdot [H^+]$$

# Learning Objectives

1. Master the basic principles of UV spectroscopy, the characteristics of various electronic transitions and absorption bands, as well as their influencing factors.
2. Be able to predict and sketch UV spectra according to compound structures.
3. Master Beer's law and its application.
4. Master quantitative methods for single- component analysis in UV spectroscopy.

# THANK YOU  FOR YOU ATTENTION!

Questions  & Discussion